Google Enterprise AI Ecosystem

Every layer of Gemini Enterprise Agent Platform and the services around it, explained in depth: models, agents, workspace, data, security and production.



BEFORE WE START

How to read this tutorial

EVERY SERVICE GETS FOUR SLIDES

Concept

What the service is, why it matters and an analogy on a sticky note.

Capabilities

Six numbered capability cards: the deep dive.

How it works

A four step flow plus three places it shines.

Field notes

Tips, pitfalls and a check yourself question with answer.

PLUS, AROUND EACH LAYER

Layer map

Every service of a layer at a glance.

Recap

Three takeaways per layer.

Quiz

Four questions with an upside down answer key.

Labs and patterns

Hands on build, architectures and checklists.

AGENDA

Nine parts, bottom of the stack to the top

00	Introduction and the big picture	*slide 1*
01	Core Models	*slide 15*
02	Agent Platform & Tools	*slide 59*
WS	Gemini across Google Workspace	*slide 95*
03	Enterprise Work & Gemini	*slide 109*
04	Data, Knowledge & Memory	*slide 145*
05	Identity, Security & Governance	*slide 181*
06	Production & Deployment	*slide 217*
07	Putting it together: labs, patterns, checklists	*slide 261*

THE HEADLINE

Vertex AI is now Gemini Enterprise Agent Platform

WHEN AND WHERE

Announced 22 April 2026 at Google Cloud Next in Las Vegas.

Google describes it as the evolution of Vertex AI: model selection, model building and agent building in one place, plus new capabilities for agent integration, DevOps, orchestration and security.

WHAT CHANGED

The renaming cheat sheet

Previous name	Name today
Vertex AI Platform	Agent Platform
Vertex AI API	Agent Platform API
Vertex AI Agent Engine	Agent Runtime
Agent Engine Sessions	Agent Platform Sessions
Vertex AI Prediction	Agent Platform Inference
Vertex AI Training	Agent Platform Managed Training

Previous name	Name today
Vertex AI Pipelines	Agent Platform Pipelines
Vertex AI Workbench	Agent Platform Workbench
Vertex AI Search	Agent Search
Gen AI evaluation service	Agent Platform Evals
Dataplex Universal Catalog	Knowledge Catalog
Google Agentspace	Gemini Enterprise app

Existing APIs, client libraries and configurations keep working; documentation and console access moved to the Agent Platform destination.

THE PLATFORM PROMISE

Four pillars: build, scale, govern, optimize

Build

Create agents code first or visually.

ADK, Agent Studio, Agent Garden, Model Garden

Scale

Run agents reliably for seconds or days.

Agent Runtime, Sessions, Memory Bank, sandboxes

Govern

Identify, control and audit every agent.

Agent Identity, Registry, Gateway, Model Armor

Optimize

Observe, evaluate and continuously improve.

Observability, Evaluation, Simulation, Optimizer

TWO FRONT DOORS

One platform, two audiences

FOR BUILDERS

Agent Platform

Developers and data scientists build, deploy and govern agents in the Google Cloud console with ADK, Agent Studio, Model Garden, Runtime and the security stack.

Layers 1, 2, 4, 5 and 6

two

FOR EVERYONE ELSE

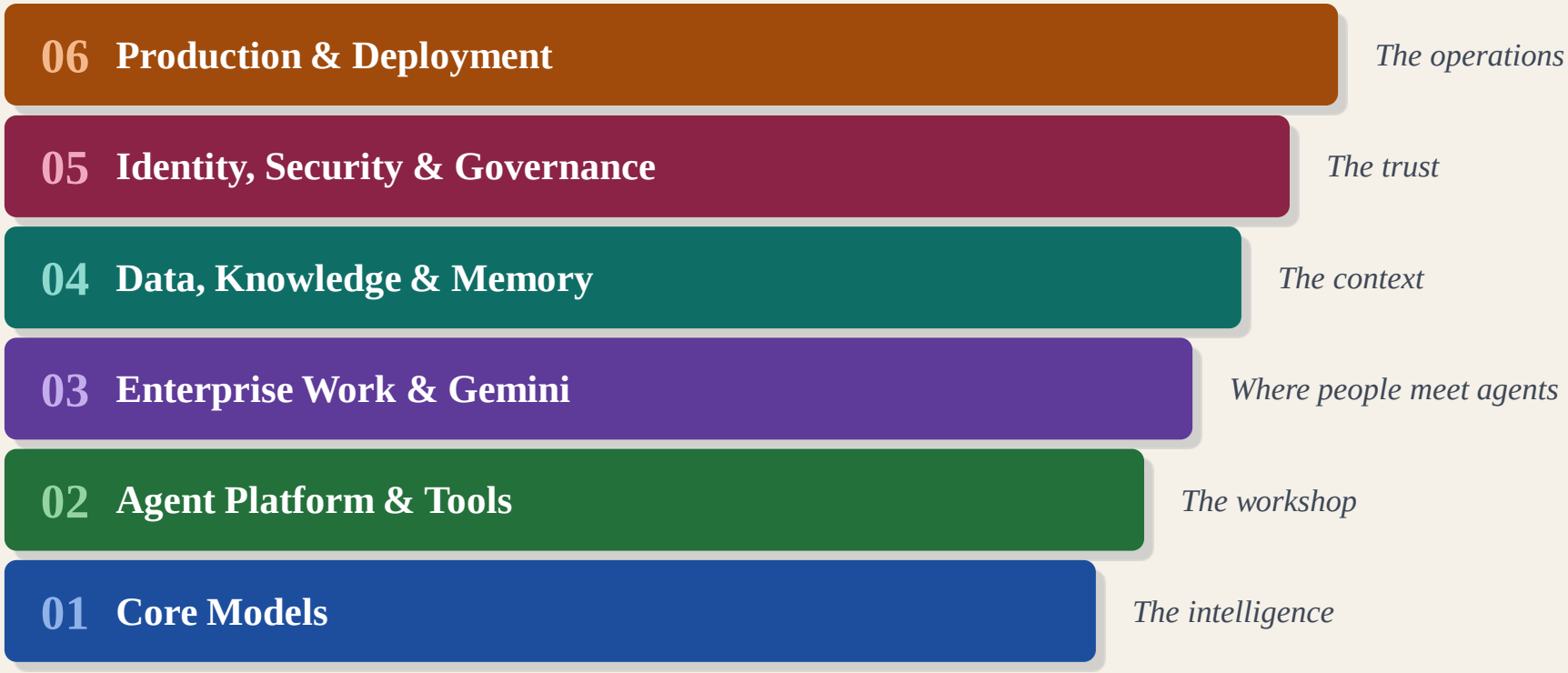
Gemini Enterprise app

Employees chat, search company data and run or build agents. It is built on Agent Platform, so identity, governance and security come standard.

Layer 3 and the Workspace band

THE ARCHITECTURE

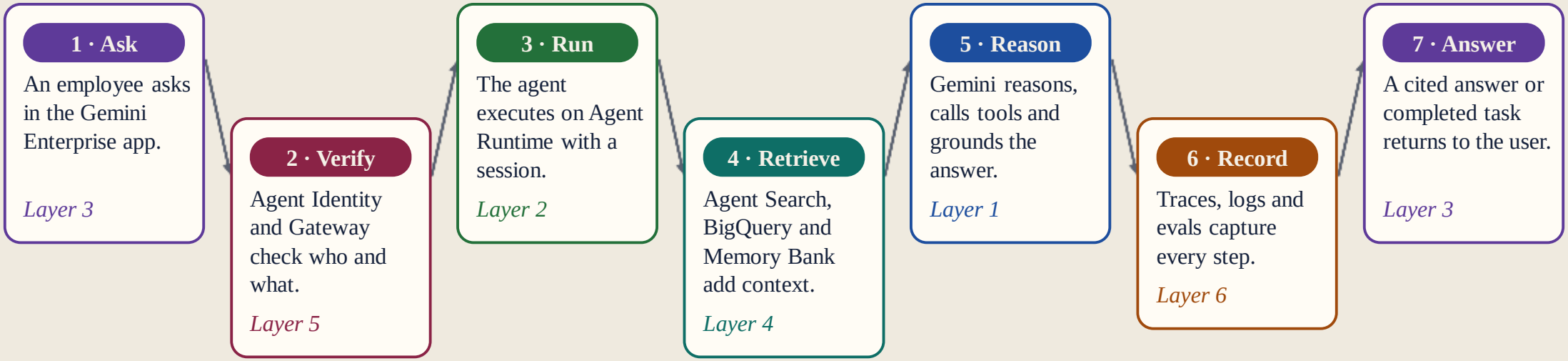
Six layers, stacked like a building



Read from the bottom up: intelligence, then the workshop, people, context, trust and operations wrap around it. The Workspace band sits between layers two and three.

A REQUEST, END TO END

How one question travels the stack



WHO USES WHAT

Personas across the six layers

Layer	Developers	Data teams	Security	Employees	Platform ops
Core Models	Primary	Uses	Reviews	Indirect	Manages
Agent Platform & Tools	Primary	Uses	Reviews	Indirect	Operates
Enterprise Work & Gemini	Uses	Light use	Reviews	Primary	Administers
Data, Knowledge & Memory	Uses	Primary	Reviews	Indirect	Operates
Identity, Security & Governance	Uses	Uses	Primary	Indirect	Primary
Production & Deployment	Primary	Uses	Uses	Indirect	Primary

Use this to decide which sections each team should study first.

VOCABULARY

Key terms, part one

Agent

Software that uses a model to reason, plan and act through tools toward a goal.

Tool

A function, API or service an agent can call to read or change the world.

Grounding

Tying answers to evidence such as search results or company documents.

RAG

Retrieval augmented generation: fetch relevant content, then generate an answer.

Embedding

A vector of numbers representing meaning, used for similarity search.

Context window

How much text, images and history a model can consider at once.

VOCABULARY

Key terms, part two

MCP

Model Context Protocol, the open standard connecting agents to tools and data.

A2A

Agent2Agent protocol, the open standard for agents to collaborate.

Session

The ordered record and state of one conversation with an agent.

Memory

Facts an agent keeps across sessions, such as user preferences.

Autorater

A judge model that scores outputs against rubrics at scale.

Sandbox

An isolated environment where agent code runs without touching real systems.

A NOTE ON CURRENCY

This ecosystem moves monthly

Preview versus GA. Many agent features launched in preview; check the release stage before committing production workloads.

Names change. Expect further renames; the cheat sheet earlier maps the big ones.

Regions differ. Memory Bank, Agent Gateway and models are not in every region.

Verify numbers. Prices, limits and model versions change; always confirm in the official docs.

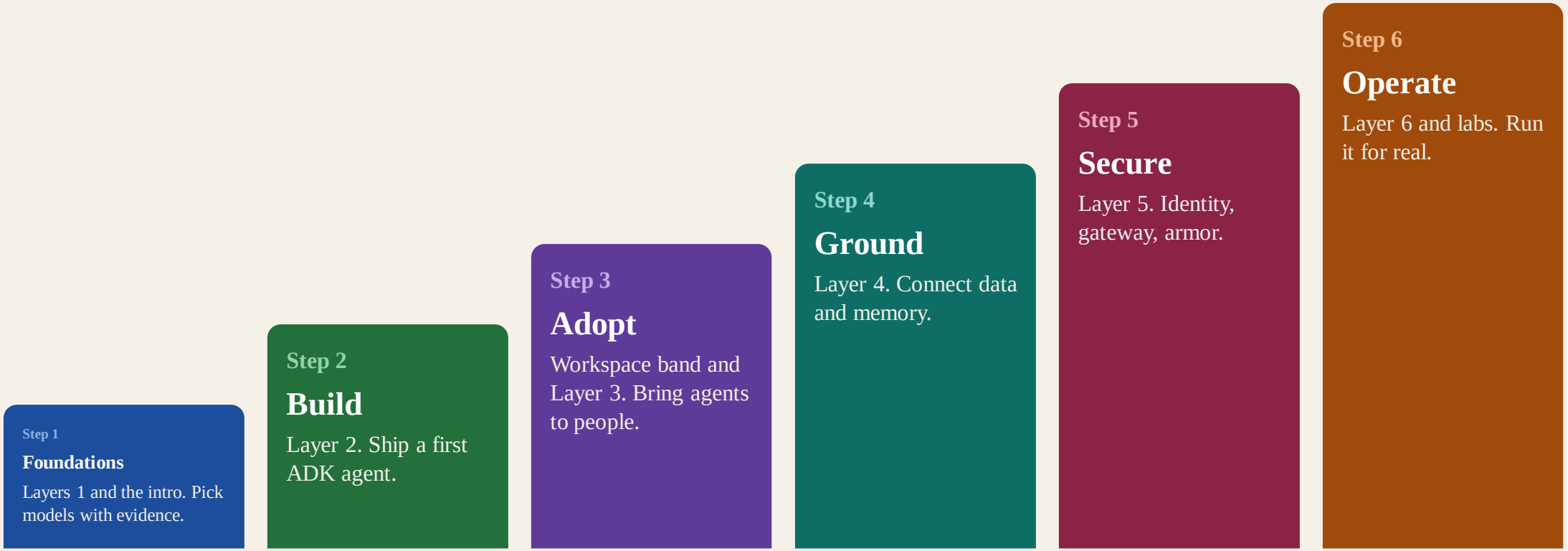
TUTORIAL DATE

October 2026

Facts reflect public announcements and documentation up to this date.

LEARNING PATH

Climb the stairs in six steps



LAYER ONE · THE INTELLIGENCE

Core Models

The brains of every agent. Google's Gemini family, generative media models, partner and open models, plus the services that test, version, ground, embed and reliably serve them at enterprise scale.

Gemini Models | Veo, Nano Banana & Lyria | Anthropic Claude | Gemma, Llama & Mistral | Gemini Embedding & Live

Model Garden | Agent Platform Evals | Model Registry | Grounding with Google Search | Provisioned Throughput

LAYER 1 MAP

The 10 services of Core Models

Gemini Models
A family of engines: a V12 for hard climbs, a hybrid for daily driving.

Veo, Nano Banana & Lyria
A full creative studio: camera crew, illustrator, composer and narrator.

Anthropic Claude
A respected guest expert who works inside your office, under your rules.

Gemma, Llama & Mistral
A well stocked library where you can borrow, annotate or own the book.

Gemini Embedding & Live
A librarian's index of meaning, and a phone line that never puts you on hold.

Model Garden
A showroom with test drives, price tags and keys to drive away.

Agent Platform Evals
A driving test with examiners, a scoring rubric and a written record.

Model Registry
A version controlled vault for every model your team builds.

Grounding with Google Search
A researcher who must footnote every claim before you see the essay.

Provisioned Throughput
Booking the whole restaurant for your event instead of hoping for a table.

CORE MODELS · SERVICE 1

Gemini Models

> *3.1 Pro for deep reasoning, Flash for speed and Flash Lite for volume, all with multimodal input and tool calling.*

WHAT IT IS

Google DeepMind's flagship model family. Each tier reads text, images, audio, video and documents, calls tools, returns structured output and works across a long context window of up to a million tokens.

WHY IT MATTERS

Most agents on the platform run on Gemini. Matching the tier to the task is the single biggest lever on quality, latency and cost.

LAYER 1 · CORE MODELS



THINK OF IT AS

A family of engines: a V12 for hard climbs, a hybrid for daily driving.

CAPABILITIES · GEMINI MODELS

DEEP DIVE

Six things Gemini gives you

01 Tiered lineup

Gemini 3.1 Pro for the hardest reasoning, Gemini 3.5 Flash for agents and coding, 3.1 Flash Lite for bulk work.

02 Native multimodality

One request can mix PDFs, screenshots, audio clips and video frames with plain text instructions.

03 Long context

Around one million tokens of context lets a model read whole contracts, codebases or meeting archives.

04 Tool calling

Function declarations let the model choose tools, fill arguments and run several calls in parallel.

05 Thinking levels

Configurable reasoning effort lets you trade extra thinking for lower latency and cost per request.

06 Structured output

JSON schema constrained responses make model output safe to parse in downstream code.

HOW IT WORKS · GEMINI MODELS

From request to result

1

Choose a tier

Start with Flash for most agents, escalate to Pro for planning and hard reasoning steps.

2

Send the request

Call the Agent Platform API with contents, system instruction, tools and generation config.

3

Model reasons

Gemini thinks at the configured level, may call tools, then composes its final answer.

4

Parse and route

Your code validates structured output, logs usage and routes follow up work to the right tier.

WHERE IT SHINES

A planning agent on Pro that delegates many small subtasks to Flash Lite workers.

Reading scanned invoices and photos together, then returning validated JSON line items.

Summarising hours of meeting video into decisions, owners and dates.

FIELD NOTES · GEMINI MODELS

Using it well in production

PRACTITIONER TIPS

1. Benchmark two tiers on your own data before committing; published leaderboards rarely match your workload.

2. Lower the thinking level for simple steps; reserve high thinking for genuinely hard decisions.

3. Use structured output plus schema validation instead of asking for JSON in the prompt.

⚠ PITFALLS TO AVOID

1. Defaulting everything to the biggest model, which inflates latency and cost for little gain.

2. Stuffing the full context window when retrieval of the right ten pages would answer better.

3. Pinning a preview model in production without a plan for its retirement or upgrade.

CHECK YOURSELF

A support bot answers simple FAQs but must occasionally plan multistep refunds. How would you tier it?

Answer: Route FAQs to Flash Lite or Flash and escalate only the refund planning step to Pro, logging both paths.

CORE MODELS · SERVICE 2

Veo, Nano Banana & Lyria

> *Veo 3.1 for video, Nano Banana for images, Lyria 3 for music and Gemini TTS for natural speech.*

WHAT IT IS

Google's generative media models. Veo creates video with sound from text or images, Nano Banana generates and edits images conversationally, Lyria composes music and Gemini TTS turns text into expressive speech.

WHY IT MATTERS

Marketing, training and product teams can produce and localise rich media inside the same governed platform that runs their agents.

LAYER 1 · CORE MODELS



THINK OF IT AS

A full creative studio: camera crew, illustrator, composer and narrator.

CAPABILITIES · VEO, NANO BANANA & LYRIA

DEEP DIVE

Six things the media models gives you

01 Video with audio

Veo produces short cinematic clips with synchronised sound from a prompt or a reference image.

02 Conversational image edits

Nano Banana edits an existing picture through plain language while keeping subjects consistent.

03 Music generation

Lyria creates instrumental tracks for ads, games and videos from mood and genre prompts.

04 Expressive speech

Gemini TTS controls voice, pace and tone, including multi speaker dialogue for podcasts.

05 Provenance

Google embeds SynthID watermarks in generated media so it can later be identified as AI made.

06 Enterprise controls

Runs under your project's IAM, quotas, regions and data governance like any other model.

HOW IT WORKS · VEO, NANO BANANA & LYRIA

From request to result

1

Brief

Describe subject, style, camera or voice; attach reference images or brand assets.

2

Generate

Call the media model; long video and music jobs run asynchronously and return an operation.

3

Review

Humans check brand fit, rights and accuracy before anything is published.

4

Iterate

Refine with follow up instructions, then store approved assets with their prompt history.

WHERE IT SHINES

- Localising one product video into twenty languages with matching narration.
- Producing on brand campaign imagery variants for A and B testing.
- Creating training scenarios with voiced dialogue for call centre onboarding.

FIELD NOTES · VEO, NANO BANANA & LYRIA

Using it well in production

PRACTITIONER TIPS

1. Keep a prompt library of approved styles so outputs stay on brand across teams.

2. Use reference images to anchor characters and products across many generations.

3. Store prompts, seeds and model versions with each asset for audit and reproduction.

⚠ PITFALLS TO AVOID

1. Publishing generated media without human review for rights, likeness and accuracy.

2. Ignoring regional availability and safety filter settings until launch day.

3. Treating media jobs like instant calls instead of polling long running operations.

CHECK YOURSELF

Why should a brand team record which model version produced each approved asset?

Answer: So assets can be reproduced, audited and traced if questions about provenance or rights arise later.

CORE MODELS · SERVICE 3

Anthropic Claude

> *Claude Opus, Sonnet and Haiku served as partner models on Agent Platform.*

WHAT IT IS

Anthropic's Claude models offered as fully managed partner models. You call them through Agent Platform with the same project, billing, IAM, quotas and regional controls you use for Gemini.

WHY IT MATTERS

Teams can pick the best model for each job, such as Claude for certain coding or writing tasks, without leaving Google Cloud governance.

LAYER 1 · CORE MODELS



THINK OF IT AS

A respected guest expert who works inside your office, under your rules.

CAPABILITIES · ANTHROPIC CLAUDE

DEEP DIVE

Six things Claude gives you

01 Three tiers

Opus for the most demanding work, Sonnet for balanced performance, Haiku for speed and cost.

02 Model as a service

No infrastructure to manage; pay per token through your Google Cloud bill.

03 Same guardrails

IAM, VPC Service Controls, audit logs and regional endpoints apply as for first party models.

04 Tool use

Claude supports tool calling, so it can power ADK agents and other frameworks.

05 Prompt caching

Repeated long prompts can be cached to cut cost and latency on supported versions.

06 Side by side evals

Compare Claude and Gemini on your own data with Agent Platform Evals.

HOW IT WORKS · ANTHROPIC CLAUDE

From request to result

1

Enable

Enable the Claude model in Model Garden and accept the partner terms for your project.

2

Call

Send requests to the regional or global endpoint using the Anthropic SDK or REST.

3

Govern

Requests inherit project IAM, logging, quotas and network perimeters.

4

Compare

Track quality and cost per model and route each task to the winner.

WHERE IT SHINES

- A coding agent that uses Claude for code review while Gemini handles search and summarisation.
- Long form drafting tasks where a team prefers Claude's writing style.
- Model diversity for resilience if one provider has a capacity issue.

FIELD NOTES · ANTHROPIC CLAUDE

Using it well in production

PRACTITIONER TIPS

1. Check regional availability and quota for each Claude version before designing around it.

2. Abstract model calls behind one interface so switching models is a config change.

3. Evaluate with identical prompts and rubrics; do not compare models on anecdotes.

⚠ PITFALLS TO AVOID

1. Assuming every Gemini only feature, such as certain grounding options, works identically for Claude.

2. Forgetting partner model pricing differs from Gemini when estimating budgets.

3. Hard coding one provider's prompt quirks throughout your codebase.

CHECK YOURSELF

What stays the same when you switch an agent from Gemini to Claude on Agent Platform?

Answer: Your project governance: IAM, audit logging, quotas, network controls and consolidated Google Cloud billing.

CORE MODELS · SERVICE 4

Gemma, Llama & Mistral

> *Open and partner models such as Gemma, Llama, Mistral, DeepSeek, Qwen, Grok and Hugging Face models.*

WHAT IT IS

A broad catalog of open weight and third party models. Some run as managed APIs; others you deploy to your own endpoints or GKE, where you control hardware, tuning and data locality.

WHY IT MATTERS

Open models give control, customisation and cost options, and they suit sovereignty, edge and fine tuning needs that closed APIs cannot meet.

LAYER 1 · CORE MODELS



THINK OF IT AS

A well stocked library where you can borrow, annotate or own the book.

CAPABILITIES · GEMMA, LLAMA & MISTRAL

DEEP DIVE

Six things open models gives you

01 Gemma family

Google's open models, built from Gemini research, sized from laptop friendly to server class.

02 Partner catalog

Llama, Mistral, DeepSeek, Qwen, Grok and more, as managed APIs or self deployed.

03 Hugging Face

Deploy thousands of Hugging Face models to managed endpoints from Model Garden.

04 Your hardware

Choose GPUs or TPUs, autoscaling and machine types for self deployed endpoints.

05 Full tuning

Fine tune open weights with your data, including parameter efficient methods like LoRA.

06 Portability

Open weights can also run on GKE, Google Distributed Cloud or on device.

HOW IT WORKS · GEMMA, LLAMA & MISTRAL

From request to result

1

Shortlist

Filter Model Garden by task, licence, size and serving option.

2

Deploy

Use one click deployment or a managed API; pick hardware and scaling.

3

Tune

Optionally fine tune on your data and register the new version.

4

Serve

Expose an endpoint, monitor latency and cost, scale with demand.

WHERE IT SHINES

- A small tuned Gemma model classifying tickets cheaply at very high volume.
- Air gapped deployments where data must never leave a sovereign environment.
- Research teams comparing open models on a specialised domain.

Using it well in production

PRACTITIONER TIPS

1. Read each model's licence; commercial terms differ between families.

2. Right size the hardware; an idle large GPU endpoint is expensive.

3. Use managed APIs first, then self deploy only when control or cost demands it.

⚠ PITFALLS TO AVOID

1. Leaving self deployed endpoints running at full scale overnight.

2. Skipping safety evaluation because a model is popular on public leaderboards.

3. Assuming an open model's tool calling format matches Gemini's.

CHECK YOURSELF

When would you choose a self deployed open model over a managed Gemini API?

Answer: When you need weight level tuning, strict data locality, offline or edge serving, or very low unit cost at scale.

CORE MODELS · SERVICE 5

Gemini Embedding & Live

> *Gemini Embedding 2 for retrieval, and Gemini Live for real time voice and video agents.*

WHAT IT IS

Two complementary services. Gemini Embedding turns content into vectors that capture meaning for search and RAG. The Live API streams audio and video both ways for natural, interruptible, real time conversations.

WHY IT MATTERS

Embeddings power accurate retrieval for every grounded agent, while Live makes voice assistants and video aware helpers practical.

LAYER 1 · CORE MODELS



THINK OF IT AS

A librarian's index of meaning, and a phone line that never puts you on hold.

DEEP DIVE

Six things embeddings and Live gives you

01 Semantic vectors

Embeddings place similar meanings close together, so search finds intent, not keywords.

02 Task types

Optimise embeddings for retrieval, similarity, classification or clustering tasks.

03 Flexible size

Configurable output dimensions balance retrieval quality against storage cost.

04 Two way streaming

Live sends and receives audio and video over a persistent low latency session.

05 Barge in

Users can interrupt mid sentence and the agent adapts, like a natural phone call.

06 Tools in real time

Live sessions can call functions and search while the conversation continues.

HOW IT WORKS · GEMINI EMBEDDING & LIVE

From request to result

1

Embed

Chunk documents and convert each chunk to a vector with Gemini Embedding.

2

Index

Store vectors in Vector Search, BigQuery or RAG Engine with metadata.

3

Converse

Open a Live session; stream microphone audio and camera frames.

4

Ground

Retrieve matching chunks and feed them to the Live agent as it answers.

WHERE IT SHINES

- A voice concierge that answers policy questions from indexed handbooks.
- A field technician pointing a phone camera at equipment for guided repair.
- Semantic deduplication of millions of support tickets.

Using it well in production

PRACTITIONER TIPS

1. Embed queries and documents with matching task types for best recall.

2. Re embed the corpus when you upgrade embedding models; vectors are not comparable across models.

3. Design Live prompts for short spoken answers, not long written paragraphs.

⚠ PITFALLS TO AVOID

1. Mixing vectors from two embedding models in one index.

2. Ignoring session limits and reconnection logic in long Live calls.

3. Sending full documents to a voice agent instead of retrieved snippets.

CHECK YOURSELF

You upgrade to a new embedding model. What must happen to your existing index?

Answer: Re embed the whole corpus with the new model, because vectors from different models are not comparable.

CORE MODELS · SERVICE 6

Model Garden

> *Discover, test and deploy Google, partner and open models from one place.*

WHAT IT IS

The catalog of more than two hundred models on Agent Platform. Each model card explains capabilities, licences and pricing, with notebooks, playground testing and one click deployment.

WHY IT MATTERS

A single governed catalog stops shadow AI, speeds evaluation and lets platform teams control which models the organisation may use.

LAYER 1 · CORE MODELS



THINK OF IT AS

A showroom with test drives, price tags and keys to drive away.

CAPABILITIES · MODEL GARDEN

DEEP DIVE

Six things Model Garden gives you

01 One catalog

Gemini, partner models like Claude, and open models from Gemma to Hugging Face in one view.

02 Model cards

Capabilities, intended use, licence, pricing and supported regions on each card.

03 Try before deploy

Prompt models in the console playground before writing any code.

04 One click deploy

Spin up an endpoint with recommended hardware and settings.

05 Org policy control

Administrators can restrict which models projects are allowed to use.

06 Notebooks

Starter notebooks show tuning, evaluation and serving for each model.

HOW IT WORKS · MODEL GARDEN

From request to result

1. **Browse**
   Filter by modality, task, provider and deployment option.
2. **Try**
   Test prompts in the playground against your sample data.
3. **Evaluate**
   Run a structured comparison with Agent Platform Evals.
4. **Deploy**
   Call the managed API or deploy to an endpoint and register it.

WHERE IT SHINES

- A platform team publishing an approved model list for all business units.
- A data scientist comparing three open models for a medical summarisation pilot.
- Quick prototyping with partner models without separate vendor contracts.

FIELD NOTES · MODEL GARDEN

Using it well in production

PRACTITIONER TIPS

1. Use organisation policies to allow list models before teams start experimenting.

2. Bookmark model cards; check them again before upgrades for changed terms.

3. Promote models from playground to evals to production through a clear gate.

⚠ PITFALLS TO AVOID

1. Deploying from the playground straight to production without evaluation.

2. Overlooking licence terms of community models.

3. Letting every team pick models with no central visibility.

CHECK YOURSELF

How can a central platform team stop teams from using unapproved models?

Answer: Use organisation policy controls on Model Garden to restrict which models projects may enable and call.

CORE MODELS · SERVICE 7

Agent Platform Evals

> *Formerly the Gen AI evaluation service: compare models and prompts before you ship.*

WHAT IT IS

A managed service that scores model and prompt outputs against datasets using computed metrics, model based autoraters and custom rubrics, either one response at a time or as pairwise comparisons.

WHY IT MATTERS

Without evaluation, prompt and model changes are guesses. Evals make quality measurable, comparable and repeatable.

LAYER 1 · CORE MODELS



THINK OF IT AS

A driving test with examiners, a scoring rubric and a written record.

CAPABILITIES · AGENT PLATFORM EVALS

DEEP DIVE

Six things Evals gives you

01 Pointwise scoring

Rate each response on fluency, groundedness, safety or your own criteria.

02 Pairwise comparison

Judge which of two models or prompts answered better, with explanations.

03 Autoraters

A judge model applies rubrics at scale, with rationales for each score.

04 Computed metrics

Exact match, ROUGE, BLEU and tool call accuracy for deterministic checks.

05 Custom rubrics

Encode your business rules, tone guides and compliance criteria as metrics.

06 Experiment tracking

Results land in Experiments so you can compare runs over time.

HOW IT WORKS · AGENT PLATFORM EVALS

From request to result

1

Build a dataset

Collect real prompts, references and edge cases, including known failures.

2

Define metrics

Combine computed metrics with rubric based autorater metrics.

3

Run

Evaluate candidate models or prompts against the dataset.

4

Decide

Compare scores, read rationales, ship the winner and keep the dataset.

WHERE IT SHINES

- Choosing between Flash and Pro for a summarisation task with evidence.
- Regression testing every prompt change in CI before deployment.
- Checking groundedness of RAG answers against source documents.

Using it well in production

PRACTITIONER TIPS

1. Calibrate autoraters against a small human labelled set before trusting them.

2. Keep a golden dataset under version control and grow it from production failures.

3. Track cost and latency alongside quality; the best answer can be too slow.

⚠ PITFALLS TO AVOID

1. Evaluating on ten hand picked prompts and calling it done.

2. Using a judge model with the same blind spots as the model being judged.

3. Optimising one metric until other qualities quietly degrade.

CHECK YOURSELF

Why calibrate an autorater against human labels?

Answer: To confirm the judge model's scores agree with expert judgement before you rely on it at scale.

CORE MODELS · SERVICE 8

Model Registry

> *Version and manage models you fine tune with supervised or reinforcement tuning.*

WHAT IT IS

The central store for your custom and tuned models. It tracks versions, aliases, lineage and evaluation results, and deploys any version to endpoints for online or batch inference.

WHY IT MATTERS

Tuned models are valuable assets. A registry makes them discoverable, reproducible, rollback ready and governed.

LAYER 1 · CORE MODELS



THINK OF IT AS

A version controlled vault for every model your team builds.

CAPABILITIES · MODEL REGISTRY

DEEP DIVE

Six things Model Registry gives you

01 Versioning

Every tuning run becomes a new version with metadata, lineage and metrics.

02 Aliases

Labels like default or champion point traffic at a version without code changes.

03 Supervised tuning

Teach a model your format and domain with labelled input and output examples.

04 Preference tuning

Reinforcement style tuning aligns outputs with human preference data.

05 Deploy anywhere

Serve a version on an endpoint, in batch, or export where supported.

06 Lineage

Trace which dataset, pipeline and base model produced each version.

HOW IT WORKS · MODEL REGISTRY

From request to result

1

Tune

Run supervised or preference tuning on a base model with your dataset.

2

Register

The tuned model lands as a new version with evaluation results.

3

Promote

Move the champion alias after evals beat the current version.

4

Roll back

If production metrics drop, point the alias back to the prior version.

WHERE IT SHINES

- A bank tuning Gemini to produce its exact credit memo format.
- Champion and challenger testing of two tuned versions.
- Audit evidence showing which model served a regulated decision.

FIELD NOTES · MODEL REGISTRY

Using it well in production

PRACTITIONER TIPS

1. Tune only after prompt engineering and retrieval have plateaued.

2. Store the training dataset snapshot reference with each version.

3. Use aliases in serving code, never hard coded version numbers.

⚠ PITFALLS TO AVOID

1. Tuning to inject facts that change often; retrieval handles that better.

2. Overfitting on a tiny dataset and celebrating training loss.

3. Losing track of which version is live in which region.

CHECK YOURSELF

Should you fine tune a model to teach it this week's product prices?

Answer: No. Prices change often, so retrieve them at runtime; tuning suits stable style, format and domain behaviour.

CORE MODELS · SERVICE 9

Grounding with Google Search

> *Plus Google Maps and your own data, so answers stay current and cited.*

WHAT IT IS

A tool that connects model answers to fresh evidence: Google Search for the public web, Google Maps for places, and enterprise sources for private data, returning citations and grounding metadata.

WHY IT MATTERS

Models alone have a training cutoff and can invent facts. Grounding reduces hallucination and gives users sources they can check.

LAYER 1 · CORE MODELS



THINK OF IT AS

A researcher who must footnote every claim before you see the essay.

DEEP DIVE

Six things Grounding gives you

01 Web grounding

Answers draw on live Google Search results instead of stale training data.

02 Maps grounding

Location aware answers use places, hours and reviews from Google Maps.

03 Your data

Ground on enterprise indexes such as Agent Search or a RAG corpus.

04 Citations

Responses include source links and supporting segments for each claim.

05 Search suggestions

Rendered search entry points let users continue research themselves.

06 Dynamic use

The model decides when grounding is needed, saving cost on simple questions.

HOW IT WORKS · GROUNDING WITH GOOGLE SEARCH

From request to result

1

Enable the tool

Add Google Search, Maps or a data store as a grounding tool.

2

Model queries

Gemini generates search queries from the user request.

3

Synthesise

Results are read and merged into one answer.

4

Cite

Grounding metadata links each claim to its source for display.

WHERE IT SHINES

- A sales assistant briefing reps on a prospect's latest public news.
- A travel agent recommending open restaurants near a hotel.
- Policy answers grounded in the internal handbook, with page citations.

Using it well in production

PRACTITIONER TIPS

1. Always render citations in the UI; grounding without visible sources wastes trust.

2. Combine web and private grounding carefully, labelling which source answered.

3. Follow Google's display requirements for search suggestions.

⚠ PITFALLS TO AVOID

1. Assuming grounding guarantees truth; sources can be wrong or outdated.

2. Grounding confidential questions on the public web.

3. Ignoring grounding cost when every trivial request triggers a search.

CHECK YOURSELF

A user asks about yesterday's regulatory change. What should the agent use?

Answer: Grounding with Google Search, so the answer reflects current information and shows citations.

CORE MODELS · SERVICE 10

Provisioned Throughput

> *Reserved capacity alongside pay as you go, batch inference and context caching.*

WHAT IT IS

A set of consumption options for serving models reliably: reserved throughput for guaranteed capacity, pay as you go for flexibility, batch for discounted offline jobs and context caching for repeated long inputs.

WHY IT MATTERS

Production agents need predictable latency and cost. The right consumption mix prevents throttling at peak and waste at quiet times.

LAYER 1 · CORE MODELS



THINK OF IT AS

Booking the whole restaurant for your event instead of hoping for a table.

DEEP DIVE

Six things Provisioned Throughput gives you

01 Reserved capacity

Buy throughput units for a model so peak traffic is served without throttling.

02 Pay as you go

Per token billing for spiky or experimental workloads beyond the reservation.

03 Spillover

Traffic above reserved capacity can overflow to pay as you go.

04 Batch inference

Send large offline jobs at a discount compared with online requests.

05 Context caching

Cache long shared prefixes like manuals so repeat calls cost less.

06 Usage monitoring

Dashboards show utilisation so you can right size reservations.

HOW IT WORKS · PROVISIONED THROUGHPUT

From request to result

1

Profile

Measure tokens per second at peak and the share of steady baseline traffic.

2

Reserve

Cover the steady baseline with Provisioned Throughput.

3

Overflow

Let spikes spill over to pay as you go; move offline work to batch.

4

Tune

Review utilisation monthly and resize the reservation.

WHERE IT SHINES

- A contact centre agent that must never throttle during Monday morning peaks.
- Nightly batch classification of millions of documents.
- Many questions over the same long policy manual using context caching.

Using it well in production

PRACTITIONER TIPS

1. Reserve for the baseline, not the absolute peak.

2. Batch anything that does not need an answer within seconds.

3. Cache long, stable prefixes and keep the variable part at the end of the prompt.

⚠ PITFALLS TO AVOID

1. Over reserving and paying for idle capacity.

2. Running bulk backfills through online endpoints at full price.

3. Placing changing content before the cached prefix, which defeats caching.

CHECK YOURSELF

Which option suits a nightly job scoring five million records?

Answer: Batch inference, because it is asynchronous, discounted and does not compete with live traffic.

LAYER 1 RECAP

Three things to remember

1. Pick models by job: Pro for hard reasoning, Flash for agentic speed, Flash Lite for volume, partner or open models when they fit better.
2. Model Garden, Model Registry and Evals turn model choice into an evidence based, versioned and repeatable engineering decision.
3. Grounding and Provisioned Throughput are what make a clever model trustworthy and dependable enough for production traffic.

LAYER 1 QUIZ

Test yourself on Core Models

QUESTION 1

Which Gemini tier would you route high volume intent classification to?

QUESTION 2

What does Grounding with Google Search add to an answer?

QUESTION 3

Name two partner model families available on Agent Platform.

QUESTION 4

What problem does Provisioned Throughput solve?

ANSWER KEY, FLIP TO READ

1. Flash Lite, the cheapest and fastest tier.

2. Fresh web facts plus citations to check them.

3. Anthropic Claude and Llama or Mistral, among others.

4. Guaranteed capacity and predictable latency at peak.

LAYER TWO · THE WORKSHOP

Agent Platform & Tools

Where developers turn models into agents. Frameworks, visual builders, templates, a managed runtime with sandboxes, and the two open protocols, A2A and MCP, that let agents reach tools and each other.

Agent Development Kit | Agent Studio | Agent Garden | Managed Agents API | Agent Runtime | Sessions & Code Execution | A2A Protocol | MCP

LAYER 2 MAP

The 8 services of Agent Platform & Tools

Agent Development Kit
A professional toolkit and blueprint set for building digital workers.

Agent Studio
A whiteboard where sketches become working machines.

Agent Garden
A seed bank of proven designs you can plant and grow.

Managed Agents API
Hiring a fully equipped contractor by filling in one work order.

Agent Runtime
A power station and control room that keeps every agent running.

Sessions & Code Execution
A notebook that remembers the conversation and a lab bench sealed behind glass.

A2A Protocol
A shared language and business card exchange for agents from any company.

MCP
A universal adapter plug for every tool socket.

AGENT PLATFORM & TOOLS · SERVICE 1

Agent Development Kit

> *Open, model agnostic framework for agents and graph based networks of sub agents.*

WHAT IT IS

Google's open source, code first framework for building, testing and deploying agents. It provides agents, tools, workflows, callbacks, sessions, memory and evaluation, with Gemini or other models.

WHY IT MATTERS

ADK gives engineers full control and the same framework Google uses internally, with a direct path to managed hosting on Agent Runtime.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A professional toolkit and blueprint set for building digital workers.

CAPABILITIES · AGENT DEVELOPMENT KIT

DEEP DIVE

Six things ADK gives you

01 Agent types

LLM agents reason freely; sequential, parallel and loop agents give deterministic workflows.

02 Graph networks

Compose sub agents into graphs so a coordinator routes work to specialists.

03 Rich tools

Python functions, built in Google tools, OpenAPI specs and MCP toolsets.

04 Callbacks

Hook before and after model and tool calls for guardrails, logging and edits.

05 Local dev UI

A web dev UI shows events, state and traces as you chat with the agent.

06 Built in eval

Evaluate trajectories and final responses against test sets in CI.

HOW IT WORKS · AGENT DEVELOPMENT KIT

From request to result

1

Define

Write an agent with a model, instruction, tools and optional sub agents.

2

Run locally

Chat in the dev UI, inspect each event, fix prompts and tools.

3

Evaluate

Run eval sets for tool use and answer quality.

4

Deploy

Ship to Agent Runtime, Cloud Run or GKE with one command.

WHERE IT SHINES

A coordinator agent routing billing, technical and sales questions to specialist sub agents.

A deterministic document pipeline: extract, validate, enrich, then file.

An internal agent that calls proprietary APIs through function tools.

Using it well in production

PRACTITIONER TIPS

1. Keep instructions short and give each sub agent one clear responsibility.

2. Use workflow agents for steps that must always happen in order.

3. Write eval sets from day one, starting with your ten most common requests.

⚠ PITFALLS TO AVOID

1. One giant agent with forty tools and a two page instruction.

2. Putting secrets in instructions or tool docstrings.

3. Testing only the happy path in the dev UI.

CHECK YOURSELF

When should you use a sequential workflow agent instead of letting an LLM agent decide?

Answer: When steps must always run in a fixed order, such as validate before submit, so behaviour stays predictable.

AGENT PLATFORM & TOOLS · SERVICE 2

Agent Studio

> *Low code visual canvas to design and test agents, then export to ADK.*

WHAT IT IS

A visual, low code builder in the Agent Platform console. You drag in models, instructions, tools and sub agents, test them in place, and export the design as ADK code when engineers take over.

WHY IT MATTERS

It lets domain experts and developers collaborate on one design, shortening the gap between idea and working prototype.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A whiteboard where sketches become working machines.

CAPABILITIES · AGENT STUDIO

DEEP DIVE

Six things Agent Studio gives you

01 Visual canvas

Lay out agents, sub agents and tools as connected blocks.

02 Instant testing

Chat with the draft agent and see tool calls as you edit.

03 Tool catalog

Attach Google tools, connectors and MCP servers without code.

04 Export to ADK

Generate ADK code so engineers can extend, test and version it.

05 Template start

Begin from Agent Garden templates instead of a blank page.

06 Shared drafts

Teams iterate on the same agent design in the console.

HOW IT WORKS · AGENT STUDIO

From request to result

1

Sketch

Place a root agent and connect specialist sub agents.

2

Equip

Add tools, grounding sources and instructions.

3

Test

Try real requests, refine prompts and tool settings.

4

Hand off

Export to ADK, commit to Git and deploy through CI.

WHERE IT SHINES

- A compliance lead designing a policy question agent with an engineer.
- Rapid prototypes for a stakeholder demo within a day.
- Teaching teams agent architecture with a visual model.

FIELD NOTES · AGENT STUDIO

Using it well in production

PRACTITIONER TIPS

1. Use Studio for discovery and design, then manage production code in Git.

2. Name agents and tools clearly; the names become part of the prompt.

3. Export early once logic stabilises to avoid drift between canvas and code.

⚠ PITFALLS TO AVOID

1. Editing in both Studio and code and losing track of the source of truth.

2. Shipping a canvas prototype without evals or security review.

3. Over nesting sub agents for a problem one agent could solve.

CHECK YOURSELF

Why export an Agent Studio design to ADK before production?

Answer: So the agent lives in version control with tests, reviews and CI, and engineers can extend it safely.

AGENT PLATFORM & TOOLS · SERVICE 3

Agent Garden

> *Prebuilt agents and templates to start from instead of a blank page.*

WHAT IT IS

A library of ready made agent samples, tools and templates covering common patterns such as customer service, data analysis, research and RAG, each deployable or customisable.

WHY IT MATTERS

Starting from a working, reviewed sample cuts weeks from delivery and teaches good patterns by example.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A seed bank of proven designs you can plant and grow.

CAPABILITIES · AGENT GARDEN

DEEP DIVE

Six things Agent Garden gives you

01 Sample agents

Complete agents for service, data science, research and more.

02 Reusable tools

Prebuilt tool implementations you can attach to your own agents.

03 Best practice

Samples show structure, evals and deployment the way Google recommends.

04 Quick deploy

Launch a sample on Agent Runtime to see it working end to end.

05 Open in Studio

Customise templates visually or in ADK code.

06 Learning path

Each sample doubles as a tutorial for a pattern.

HOW IT WORKS · AGENT GARDEN

From request to result

1

Browse

Find a template close to your use case.

2

Deploy

Run it unchanged to see the baseline behaviour.

3

Customise

Swap in your data, tools, instructions and branding.

4

Harden

Add evals, identity and gateway policies before launch.

WHERE IT SHINES

- A retailer adapting a customer service sample for order tracking.
- A data team starting from a SQL analysis agent sample.
- Workshops where teams deploy a template in their first hour.

FIELD NOTES · AGENT GARDEN

Using it well in production

PRACTITIONER TIPS

1. Read the sample's README and evals before changing anything.

2. Keep a fork so you can pull upstream improvements.

3. Replace demo credentials and data stores immediately.

⚠ PITFALLS TO AVOID

1. Shipping a demo template to customers as is.

2. Assuming sample prompts suit your domain language.

3. Ignoring the security notes that come with samples.

CHECK YOURSELF

What is the first hardening step after deploying an Agent Garden sample?

Answer: Replace demo data and credentials, then add evals plus identity and gateway policies before real users touch it.

AGENT PLATFORM & TOOLS · SERVICE 4

Managed Agents API

> *Config driven REST API for autonomous agents inside a managed sandbox.*

WHAT IT IS

A declarative API where you describe an agent's model, instructions, tools and limits in configuration and Google runs it as an autonomous agent inside a managed, isolated sandbox.

WHY IT MATTERS

Teams get autonomous agents without writing orchestration code or operating infrastructure, which suits product teams and quick integrations.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

Hiring a fully equipped contractor by filling in one work order.

DEEP DIVE

Six things Managed Agents API gives you

01 Config not code

Define the agent in a declarative configuration sent over REST.

02 Managed sandbox

Each agent runs in isolated compute with controlled file and network access.

03 Autonomy

Agents plan, call tools and iterate toward a goal until done or limits hit.

04 Limits

Set budgets for steps, time and tool calls to bound behaviour.

05 Any client

Call from any language or system that speaks HTTP.

06 Platform governance

Identity, gateway policies and observability still apply.

HOW IT WORKS · MANAGED AGENTS API

From request to result

1

Configure

Write the agent configuration with model, tools and limits.

2

Create

POST the configuration to create a managed agent.

3

Run

Send a task; the agent works inside its sandbox.

4

Collect

Poll or stream progress and fetch results and artifacts.

WHERE IT SHINES

- A SaaS product adding a background research agent via one REST call.
- Automated report generation in a locked down sandbox.
- Integrating agents into legacy systems that cannot host Python.

Using it well in production

PRACTITIONER TIPS

1. Start with tight step and time limits, then relax them with evidence.
2. Keep configurations in version control like any other code.
3. Log every task's inputs and outputs for audit.

⚠ PITFALLS TO AVOID

1. Granting broad network access to an autonomous sandbox.
2. Running open ended goals without budgets.
3. Treating config agents as exempt from evaluation.

CHECK YOURSELF

Why set explicit step and time limits on an autonomous managed agent?

Answer: To bound cost and blast radius, ensuring a confused agent stops rather than looping indefinitely.

AGENT PLATFORM & TOOLS · SERVICE 5

Agent Runtime

> *Managed runtime with sub second cold starts and agents that run for days.*

WHAT IT IS

Managed hosting for agents, formerly Agent Engine. It deploys agents built with ADK and other frameworks, scales them automatically and connects them to Sessions, Memory Bank, sandboxes and observability.

WHY IT MATTERS

Running agents reliably needs scaling, state, security and monitoring. Runtime provides all of that so teams focus on agent logic.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A power station and control room that keeps every agent running.

CAPABILITIES · AGENT RUNTIME

DEEP DIVE

Six things Agent Runtime gives you

01 Fast start

Google cites sub second cold starts, so idle agents wake quickly.

02 Long running

Agents can work on tasks for hours or days, not just single requests.

03 Many frameworks

Hosts ADK, LangGraph, LangChain, LlamaIndex, AG2, A2A and custom agents.

04 Autoscaling

Scales instances with traffic and down when quiet.

05 Integrated state

Built in Sessions and Memory Bank keep context across turns.

06 Enterprise security

Supports VPC Service Controls, CMEK and compliance controls.

HOW IT WORKS · AGENT RUNTIME

From request to result

1

Package

Wrap your agent with its requirements and configuration.

2

Deploy

Push to Runtime in a chosen region; get a resource name.

3

Invoke

Apps query the agent; Runtime routes, scales and stores state.

4

Operate

Watch traces, logs and metrics; roll new versions.

WHERE IT SHINES

- A multi day procurement agent waiting for approvals and resuming work.
- Hosting a LangGraph agent without managing Kubernetes.
- Serving thousands of concurrent customer conversations.

FIELD NOTES · AGENT RUNTIME

Using it well in production

PRACTITIONER TIPS

1. Pick the region next to your data and users for latency and residency.

2. Keep agents stateless in code; let Sessions and Memory hold state.

3. Version deployments so you can roll back quickly.

⚠ PITFALLS TO AVOID

1. Storing conversation state in global variables inside the agent.

2. Ignoring regional availability of Memory Bank or Gateway.

3. Deploying without observability enabled.

CHECK YOURSELF

Why should agent code avoid storing state in memory variables when hosted on Runtime?

Answer: Instances scale and restart, so state must live in Sessions or Memory Bank to survive.

AGENT PLATFORM & TOOLS · SERVICE 6

Sessions & Code Execution

> *Conversation state, plus a sandbox that runs model written code safely.*

WHAT IT IS

Sessions store the ordered events and state of each conversation. Code Execution provides an isolated, managed sandbox where agents run generated code, shell commands or computer use without touching your systems.

WHY IT MATTERS

Agents need memory of the current task and a safe place to compute. These services provide both without custom infrastructure.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A notebook that remembers the conversation and a lab bench sealed behind glass.

CAPABILITIES · SESSIONS & CODE EXECUTION

DEEP DIVE

Six things Sessions and sandboxes gives you

01 Event history

Each session records user messages, model turns and tool calls in order.

02 Session state

Key value state lets agents track progress, choices and variables.

03 Resumable

Users return later and the agent resumes the same session.

04 Isolated code

Model written Python runs in a sandbox, not on your servers.

05 File handling

Sandboxes accept uploads and return charts, files and results.

06 Computer use

Sandboxed browsers and desktops let agents operate interfaces safely.

HOW IT WORKS · SESSIONS & CODE EXECUTION

From request to result

1

Open session

Create a session per user conversation with a user ID.

2

Append events

Every turn and tool call is written to the session.

3

Compute

When maths or data work is needed, the agent sends code to the sandbox.

4

Return

Results flow back into the session and the answer.

WHERE IT SHINES

- A finance agent running pandas analysis on an uploaded spreadsheet.
- Multi turn onboarding that remembers which steps are complete.
- Generating charts from query results without exposing servers.

Using it well in production

PRACTITIONER TIPS

1. Scope sessions per user and per task to avoid context bleed.
2. Set expiry on sessions that hold sensitive data.
3. Return compact results from sandboxes, not huge raw dumps.

⚠ PITFALLS TO AVOID

1. Executing model generated code directly on application servers.
2. Letting one user's session ID be guessed or reused by another.
3. Allowing sandboxes outbound internet access by default.

CHECK YOURSELF

Why run model generated code in a sandbox instead of your app server?

Answer: Generated code can be wrong or manipulated; isolation keeps mistakes and attacks away from real systems.

AGENT PLATFORM & TOOLS · SERVICE 7

A2A Protocol

> *Agent2Agent: an open standard for agents to discover and delegate to each other.*

WHAT IT IS

An open protocol, governed by the Linux Foundation, that lets agents built on different frameworks and vendors discover each other's skills, delegate tasks, exchange messages and artifacts and stream progress.

WHY IT MATTERS

Enterprises will run agents from many vendors. A2A lets them cooperate without custom point to point integrations.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A shared language and business card exchange for agents from any company.

CAPABILITIES · A2A PROTOCOL

DEEP DIVE

Six things A2A gives you

01 Agent card

Each agent publishes a JSON card describing skills, endpoint and auth needs.

02 Tasks

Work is a task with a lifecycle: submitted, working, input required, completed.

03 Messages and parts

Exchange text, files and structured data as typed parts.

04 Artifacts

Agents return outputs as artifacts the caller can use.

05 Streaming

Server sent events stream progress on long tasks.

06 Opaque agents

Agents collaborate without exposing internal prompts or tools.

HOW IT WORKS · A2A PROTOCOL

From request to result

1

Discover

The client agent fetches the remote agent card from a well known URL.

2

Authenticate

It obtains credentials using the scheme the card declares.

3

Delegate

It sends a task message and receives a task ID.

4

Collect

It streams updates and receives artifacts when done.

WHERE IT SHINES

- An HR agent asking a payroll vendor's agent to process a change.
- A travel agent delegating to airline and hotel agents.
- Linking ADK agents to partner agents in the Agent Gallery.

FIELD NOTES · A2A PROTOCOL

Using it well in production

PRACTITIONER TIPS

1. Write precise skill descriptions in the agent card; they drive discovery.

2. Use Agent Gateway to enforce policy on A2A traffic.

3. Design for input required states so humans can step in.

⚠ PITFALLS TO AVOID

1. Exposing internal agents publicly without authentication.

2. Trusting remote agent output without validation.

3. Confusing A2A, agent to agent, with MCP, agent to tool.

CHECK YOURSELF

What is the difference between A2A and MCP?

Answer: A2A connects agents to other agents as peers; MCP connects an agent to tools, data and resources.

AGENT PLATFORM & TOOLS · SERVICE 8

MCP

> *Model Context Protocol: a standard way for agents to reach tools and data.*

WHAT IT IS

An open standard, introduced by Anthropic and widely adopted, for connecting AI applications to tools, data and prompts through MCP servers. Google offers managed MCP servers for services such as BigQuery.

WHY IT MATTERS

Build a tool once as an MCP server and every compliant agent can use it, which cuts integration work dramatically.

LAYER 2 · AGENT PLATFORM



THINK OF IT AS

A universal adapter plug for every tool socket.

DEEP DIVE

Six things MCP gives you

01 Tools

Servers expose callable functions with JSON schema inputs.

02 Resources

Servers share readable data such as files, records or documents.

03 Prompts

Servers provide reusable prompt templates for common tasks.

04 Transports

Local stdio or remote streamable HTTP connections.

05 Managed servers

Google hosts MCP servers for services like BigQuery, AlloyDB and Cloud SQL.

06 Ecosystem

Thousands of community and vendor servers for SaaS tools.

HOW IT WORKS · MCP

From request to result

1

Connect

The agent's MCP client connects to a server and lists tools.

2

Choose

The model selects a tool based on its description and schema.

3

Call

The client sends arguments; the server performs the action.

4

Use

Results return to the model, which continues reasoning.

WHERE IT SHINES

- An ADK agent querying BigQuery through the managed MCP server.
- Bringing your own MCP server for an internal ticketing system.
- Coding agents reading docs and running tests through MCP tools.

FIELD NOTES · MCP

Using it well in production

PRACTITIONER TIPS

1. Give tools short, unambiguous names and descriptions.

2. Register approved MCP servers in Agent Registry.

3. Route MCP traffic through Agent Gateway with Model Armor checks.

⚠ PITFALLS TO AVOID

1. Installing unvetted community servers with broad permissions.

2. Tool descriptions that can be poisoned with hidden instructions.

3. Exposing write tools without human confirmation.

CHECK YOURSELF

Why route MCP traffic through Agent Gateway?

Answer: To enforce access policy, inspect for tool poisoning and prompt injection, and log every agent to tool call.

LAYER 2 RECAP

Three things to remember

1. Build code first with ADK or visually with Agent Studio; both land on the same managed Agent Runtime.
2. Runtime, Sessions, Memory and sandboxed Code Execution remove the undifferentiated plumbing of running agents for hours or days.
3. MCP connects agents to tools and data; A2A connects agents to other agents. Open protocols prevent lock in.

LAYER 2 QUIZ

Test yourself on Agent Platform & Tools

QUESTION 1

Which protocol lets an agent discover and delegate work to another agent?

QUESTION 2

Where would a business analyst start building an agent visually?

QUESTION 3

What does the Code Execution sandbox protect against?

QUESTION 4

Name three frameworks Agent Runtime can host.

ANSWER KEY, FLIP TO READ

1. A2A, the Agent2Agent protocol.
2. Agent Studio, then export to ADK if needed.
3. Model written code touching your real systems.
4. ADK, LangGraph, LangChain, LlamaIndex or AG2.

THE WORKSPACE BAND

Gemini embedded across Google Workspace

Before building custom agents, most employees meet AI inside the apps they already use. This band sits between the agent platform and the enterprise work layer on the infographic.

Gmail

AI Inbox and drafting

Docs

Writing and infographics

Sheets

Analysis in plain language

Slides

Decks from your content

Meet

Take notes for me

Chat & Drive

Summaries and answers

WORKSPACE BAND · GMAIL

Gemini in Gmail: AI Inbox and drafting

Gmail

Gemini in Gmail prioritises what needs you, summarises long threads, drafts replies in your voice and searches your inbox in plain language.

AI Inbox

Surfaces priority messages and tasks based on who and what matters to you.

Help me write

Drafts and refines emails from a short instruction.

Thread summaries

Condenses long conversations into decisions and asks.

BEFORE

Forty unread threads, ten minutes before a meeting.

AFTER

A ranked list of what needs you, with drafts ready for the top three.

TRY IT · GMAIL

Prompts that work in Gmail

Summarise every thread from the Lagos supplier this week and list open questions.

Draft a polite reply declining Thursday, proposing Monday at ten instead.

A bullet summary with owners and dates, then a short draft you edit and send.

ADMIN NOTE

Admins control Gemini features per organisational unit; AI Inbox availability depends on edition.

Docs

Gemini in Docs: Writing and infographics

Gemini in Docs drafts, rewrites, summarises and restructures documents, pulls in content from Drive files, and can create visuals such as infographics from your text.

Help me create

Generate a first draft grounded on referenced Drive files.

Refine

Shorten, formalise, simplify or change tone of any selection.

Visuals

Turn sections into images and infographic style visuals.

BEFORE

A blank page and six source documents to synthesise.

AFTER

A structured first draft citing the sources, ready for expert editing.

TRY IT · DOCS

Prompts that work in Docs

Using the three linked research notes, draft a two page brief with an executive summary.

Rewrite this section for a non technical board audience in under 150 words.

A draft brief with headings and a tighter board ready section.

ADMIN NOTE

Drafts use only files the author can access; review facts before sharing.

Sheets

Gemini in Sheets: Analysis in plain language

Gemini in Sheets builds tables, formulas, pivots and charts from natural language, explains trends, and can apply AI to cells to classify or summarise text at scale.

Ask your data

Plain language questions return answers, charts and pivots.

Formula help

Generates and explains complex formulas.

AI in cells

Classify, extract or summarise text across rows.

BEFORE

A raw export of 5,000 survey rows and a deadline.

AFTER

Themes tagged per row, a pivot by region and a chart for the deck.

TRY IT · SHEETS

Prompts that work in Sheets

Tag each comment in column C as praise, complaint or request.

Which region had the largest month on month drop in sales, and why might that be?

A new column of tags plus a chart and short explanation.

ADMIN NOTE

Validate AI generated formulas and tags on a sample before relying on them.

WORKSPACE BAND · SLIDES

Gemini in Slides: Decks from your content

Slides

Gemini in Slides generates slides, layouts and images from prompts or from referenced Docs and Sheets, and helps rewrite and restyle existing presentations.

Generate slides

Create new slides from a description or a source file.

Images

Generate on brand images for slides.

Rewrite

Tighten speaker friendly text across slides.

BEFORE
A finished report that still needs a ten slide summary.

→

AFTER
A draft deck pulling key points and charts from the report.

TRY IT · SLIDES

Prompts that work in Slides

Create five slides summarising the linked quarterly report for the sales team.

Generate a clean illustration of a warehouse robot for the title slide.

Draft slides with headings and points, plus an image to place.

ADMIN NOTE

Keep brand templates applied; review every generated claim and number.

WORKSPACE BAND · MEET

Gemini in Meet: Take notes for me

Meet

Gemini in Meet takes notes, captures decisions and action items, translates captions and lets late joiners catch up on what they missed.

Take notes for me

Automatic notes saved to Docs and shared with attendees.

Catch up

Summaries of what happened before you joined.

Translated captions

Live captions across many languages.

BEFORE

Someone types notes and misses half the discussion.

AFTER

Everyone participates; notes with owners arrive in Docs automatically.

TRY IT · MEET

Prompts that work in Meet

Summarise what I missed in the first fifteen minutes.

List every action item with its owner from today's meeting.

A catch up summary in the meeting, then a notes Doc with actions.

ADMIN NOTE

Tell participants when notes are on; follow consent and retention rules.

WORKSPACE BAND · CHAT & DRIVE

Gemini in Chat & Drive: Summaries and answers

Chat & Drive

Gemini in Chat summarises busy spaces and answers questions; in Drive it summarises files and folders and answers questions across documents you can access.

Space summaries

Catch up on long Chat spaces in seconds.

File summaries

Overviews of documents and PDFs in Drive.

Ask across files

Questions answered across many files with links.

BEFORE
Three hundred unread Chat messages and a folder of forty PDFs.

AFTER
A short catch up and a cited answer from the right PDFs.

TRY IT · CHAT & DRIVE

Prompts that work in Chat & Drive

Summarise the project space since Monday and list blockers.

Across this folder, what are the payment terms with each supplier?

A blocker list and a table of terms linking each source file.

ADMIN NOTE

Answers respect Drive sharing, so fix oversharing before rollout.

WORKSPACE BAND QUIZ

Test yourself on Gemini in Workspace

QUESTION 1

Which Gmail feature ranks the messages that need you?

QUESTION 2

What does Take notes for me produce?

QUESTION 3

Do Drive answers ignore sharing permissions?

QUESTION 4

Which app tags thousands of rows in plain language?

ANSWER KEY, FLIP TO READ

1. AI Inbox.
2. A notes Doc with decisions and actions.
3. No, they respect Drive sharing.
4. Sheets, using AI in cells.

LAYER THREE · WHERE PEOPLE MEET AGENTS

Enterprise Work & Gemini

The employee facing layer. The Gemini Enterprise app is the front door; no code builders, galleries, research agents, shared projects, connectors and an agent first IDE bring agents into everyday work.

- Gemini Enterprise App
- Agent Designer & Skills
- Agent Gallery
- Deep Research & Data Insights
- Workspace Studio
- Projects, Canvas & Inbox
- Connectors & BYO MCP
- Antigravity

LAYER 3 MAP

The 8 services of Enterprise Work & Gemini

Gemini Enterprise App

The reception desk of a smart office that knows every room and file.

Agent Designer & Skills

Describing the job to a capable assistant who then writes their own checklist.

Agent Gallery

An app store for agents, with security checks at the door.

Deep Research & Data Insights

A research analyst who reads everything overnight and briefs you at nine.

Workspace Studio

A personal assistant who watches your inbox and acts on your rules.

Projects, Canvas & Inbox

A project war room where humans and agents share the same whiteboard.

Connectors & BYO MCP

Keys to every filing cabinet, each opening only the drawers you may see.

Antigravity

A cockpit where you direct a crew of coding agents instead of typing every line.

ENTERPRISE WORK & GEMINI · SERVICE 1

Gemini Enterprise App

> *The front door for employees: chat, search and run agents across company data.*

WHAT IT IS

The employee experience of Gemini Enterprise, the evolution of Agentspace. Staff chat with Gemini, search across connected company data, and discover, create, share and run agents in one governed interface.

WHY IT MATTERS

It brings agents to every employee with identity, permissions and governance built in, replacing scattered AI tools.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

The reception desk of a smart office that knows every room and file.

CAPABILITIES · GEMINI ENTERPRISE APP

DEEP DIVE

Six things the Gemini Enterprise app gives you

01 Unified search

Permission aware search across Workspace, Microsoft 365 and business apps.

02 Chat with context

Gemini answers using company data the user is allowed to see.

03 Agent hub

Find, run and share agents from Google, colleagues and partners.

04 Built on Agent Platform

Inherits identity, governance and security from the platform.

05 Admin controls

IT manages connectors, agents, permissions and audit centrally.

06 Editions

Business and enterprise editions scale from small teams to global firms.

HOW IT WORKS · GEMINI ENTERPRISE APP

From request to result

1

Sign in

Employees authenticate with corporate identity.

2

Ask

Type a question or pick an agent from the gallery.

3

Retrieve

Connectors fetch only content the user may access.

4

Act

Gemini answers with citations or an agent completes the task.

WHERE IT SHINES

- A new hire asking where the travel policy lives and how to book.
- Sales reps prepping accounts with CRM and email context in one place.
- Managers running an approved expense review agent.

FIELD NOTES · GEMINI ENTERPRISE APP

Using it well in production

PRACTITIONER TIPS

1. Start with high value connectors such as Drive, SharePoint and CRM.

2. Publish a small set of trusted agents before opening creation to all.

3. Train staff to check citations, not just answers.

⚠ PITFALLS TO AVOID

1. Connecting every source on day one with messy permissions.

2. Assuming search fixes over shared files; it reveals them.

3. No owner for agent quality inside the app.

CHECK YOURSELF

Why audit file sharing permissions before connecting sources to the Gemini Enterprise app?

Answer: Search is permission aware, so oversharing becomes instantly discoverable; fix access first.

ENTERPRISE WORK & GEMINI · SERVICE 2

Agent Designer & Skills

> *No code agent builder with natural language, visual flows and reusable Skills.*

WHAT IT IS

A builder inside the Gemini Enterprise app where employees describe the agent they want in plain language, refine it in a visual flow, and package repeatable know how as Skills that agents reuse.

WHY IT MATTERS

Domain experts know the work best. Agent Designer lets them automate it without waiting in an engineering queue.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

Describing the job to a capable assistant who then writes their own checklist.

DEEP DIVE

Six things Agent Designer gives you

01 Natural language start

Describe the goal and Gemini drafts the agent's steps and tools.

02 Visual flows

Adjust steps, branches and approvals on a simple canvas.

03 Skills

Package instructions, templates and tools as reusable capabilities.

04 Company context

Agents use connected data with the builder's permissions model.

05 Sharing

Share agents with teammates or publish to the gallery after review.

06 Governed

Admins control who can build, publish and use agents.

HOW IT WORKS · AGENT DESIGNER & SKILLS

From request to result

1

Describe

Explain the task, inputs, outputs and data sources.

2

Refine

Edit the generated flow and add Skills.

3

Test

Run on real examples and adjust instructions.

4

Share

Publish to a team or submit to the Agent Gallery.

WHERE IT SHINES

An HR partner building a policy question agent.

A finance analyst automating monthly variance commentary.

A marketing Skill for writing in brand voice reused by many agents.

Using it well in production

PRACTITIONER TIPS

1. Write Skills for knowledge that repeats across agents, like brand voice.
2. Test with real messy inputs, not tidy examples.
3. Name an owner for every shared agent.

⚠ PITFALLS TO AVOID

1. Publishing personal experiments to the whole company.
2. Hiding critical business rules inside one unreviewed agent.
3. Duplicating the same Skill in many agents.

CHECK YOURSELF

What is the benefit of a Skill over copying instructions into each agent?

Answer: A Skill is written once, improved once and reused everywhere, keeping behaviour consistent.

ENTERPRISE WORK & GEMINI · SERVICE 3

Agent Gallery

> *Google, in house and partner agents in one place, with Agent Marketplace.*

WHAT IT IS

The catalog inside the Gemini Enterprise app where employees discover agents built by Google, by their own company, and by partners from Google Cloud Marketplace such as Adobe, Atlassian, ServiceNow and Workday.

WHY IT MATTERS

A curated gallery spreads proven agents quickly and keeps procurement, security validation and access control in one place.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

An app store for agents, with security checks at the door.

DEEP DIVE

Six things the Agent Gallery gives you

01 Three sources

Google made, company built and partner agents side by side.

02 Marketplace

Partner agents purchasable through Google Cloud Marketplace.

03 Validated

Google validates partner agents for security and interoperability.

04 Access control

Admins choose which agents appear for which groups.

05 A2A interoperability

Partner agents can collaborate with your agents via A2A.

06 Discovery

Search and categories help staff find the right agent.

HOW IT WORKS · AGENT GALLERY

From request to result

1. **Curate**
   Admins approve agents and assign them to groups.
2. **Discover**
   Employees browse or search the gallery.
3. **Use**
   Run the agent with company context and permissions.
4. **Review**
   Track usage and feedback; retire weak agents.

WHERE IT SHINES

- Adding a partner ITSM agent for ticket handling.
- Publishing an internal contract review agent company wide.
- Department specific galleries for finance and legal.

FIELD NOTES · AGENT GALLERY

Using it well in production

PRACTITIONER TIPS

1. Keep the gallery small and high quality at launch.

2. Show owner, purpose and data used for each agent.

3. Review usage quarterly and remove unused agents.

⚠ PITFALLS TO AVOID

1. Flooding the gallery with near duplicate agents.

2. Enabling partner agents without data processing review.

3. No feedback loop from users to agent owners.

CHECK YOURSELF

Who validates partner agents listed for the Gemini Enterprise app?

Answer: Google Cloud validates them for security and interoperability, and your admins still decide who may use them.

ENTERPRISE WORK & GEMINI · SERVICE 4

Deep Research & Data Insights

> *Google built agents for research across web and internal data.*

WHAT IT IS

Prebuilt Google agents. Deep Research plans multistep investigations, reads many sources and writes cited reports. Data Insights style agents answer questions over enterprise data in natural language.

WHY IT MATTERS

Research and analysis consume huge expert time. These agents produce first drafts in minutes, which experts verify and refine.

LAYER 3 · ENTERPRISE WORK



THINK OF IT AS

A research analyst who reads everything overnight and briefs you at nine.

CAPABILITIES · DEEP RESEARCH & DATA INSIGHTS

DEEP DIVE

Six things the research agents gives you

01 Research planning

Breaks a question into a plan and shows it for approval.

02 Many sources

Reads web pages and internal documents in one investigation.

03 Cited reports

Produces structured reports with references.

04 Data questions

Natural language questions over tables with charts and SQL.

05 Knowledge Catalog

Uses governed business context to interpret data correctly.

06 Export

Reports flow into Docs and shared projects.

HOW IT WORKS · DEEP RESEARCH & DATA INSIGHTS

From request to result

1

Ask

Pose a complex question with scope and sources.

2

Plan

Review and adjust the proposed research plan.

3

Investigate

The agent searches, reads and analyses iteratively.

4

Report

Receive a cited report or data answer to verify.

WHERE IT SHINES

- Competitive landscape briefs before a strategy offsite.
- Analysts asking why churn rose last quarter by region.
- Regulatory scans across many jurisdictions.

Using it well in production

PRACTITIONER TIPS

1. Edit the research plan; scope is the biggest quality lever.
2. Verify key claims against citations before sharing.
3. Point data agents at curated, documented tables.

⚠ PITFALLS TO AVOID

1. Forwarding reports unread as final work.
2. Running data questions over undocumented raw tables.
3. Asking vague questions and blaming vague answers.

CHECK YOURSELF

What is the most effective way to improve a Deep Research report?

Answer: Review and tighten the research plan before it runs, clarifying scope, sources and the decision it supports.

ENTERPRISE WORK & GEMINI · SERVICE 5

Workspace Studio

> *No code agents and skills that automate work inside Gmail, Docs and Sheets.*

WHAT IT IS

Formerly Workspace Flows, Workspace Studio lets users describe automations in plain language, and Gemini builds agents that trigger on events in Gmail, Drive, Sheets and Chat, reason over content and take actions.

WHY IT MATTERS

It moves automation beyond rigid if this then that rules, so everyone can delegate repetitive Workspace tasks.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

A personal assistant who watches your inbox and acts on your rules.

DEEP DIVE

Six things Workspace Studio gives you

01 Plain language build

Describe the automation and Gemini creates the agent.

02 Triggers

Start on new emails, form responses, schedules or file changes.

03 Reasoning steps

Gemini decides, summarises, extracts and drafts within steps.

04 Workspace actions

Label mail, update Sheets, post to Chat, create Docs.

05 Extensibility

Third party apps, webhooks and Apps Script custom steps.

06 Sharing

Share agents with teammates like sharing a Doc.

HOW IT WORKS · WORKSPACE STUDIO

From request to result

1

Describe

Say what to automate, such as flag emails with questions.

2

Review

Check the generated trigger, steps and actions.

3

Activate

Turn it on; it runs in the background.

4

Monitor

Check activity history and refine.

WHERE IT SHINES

- Labelling emails that contain direct questions and pinging you in Chat.
- Logging invoice details from attachments into a Sheet.
- Friday reminders to update a project tracker.

FIELD NOTES · WORKSPACE STUDIO

Using it well in production

PRACTITIONER TIPS

1. Start with one narrow automation and expand.
2. Use Apps Script steps for internal systems.
3. Review activity logs during the first week.

⚠ PITFALLS TO AVOID

1. Automations that send external emails without review.
2. Overlapping agents acting on the same messages.
3. Forgetting admin settings for sharing and webhooks.

CHECK YOURSELF

What makes a Workspace Studio agent different from a classic rule based filter?

Answer: It uses Gemini to understand content and intent, so it adapts to wording rather than matching fixed rules.

ENTERPRISE WORK & GEMINI · SERVICE 6

Projects, Canvas & Inbox

> *Shared spaces for teams and agents, plus one inbox to manage agent activity.*

WHAT IT IS

Collaboration surfaces in the Gemini Enterprise app. Projects group files, chats and agents around a goal, Canvas is a shared editing surface, and Inbox collects agent updates, approvals and requests in one queue.

WHY IT MATTERS

As agents do more work, people need one place to supervise them, approve actions and collaborate on outputs.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

A project war room where humans and agents share the same whiteboard.

DEEP DIVE

Six things Projects, Canvas and Inbox gives you

01 Projects

Group sources, conversations and agents around one goal.

02 Shared context

Everyone, including agents, works from the same project sources.

03 Canvas

Co edit documents and code with Gemini in a shared surface.

04 Inbox

A single queue of agent results, questions and approval requests.

05 Human in the loop

Approve or reject agent actions before they execute.

06 Activity history

See what each agent did and when.

HOW IT WORKS · PROJECTS, CANVAS & INBOX

From request to result

1

Create project

Add goal, members, files and agents.

2

Delegate

Assign tasks to agents within the project.

3

Supervise

Approve requests and review outputs in Inbox.

4

Finish

Refine in Canvas and publish the deliverable.

WHERE IT SHINES

- A bid team with agents drafting sections while humans review in Canvas.
- An approvals queue for purchase orders proposed by agents.
- A research project with shared sources and briefs.

Using it well in production

PRACTITIONER TIPS

1. Keep project sources curated; agents inherit them as context.

2. Use Inbox approvals for any irreversible action.

3. Archive projects when done to limit stale context.

⚠ PITFALLS TO AVOID

1. Letting agents act without an approval step on money or legal actions.

2. Dumping every file into one project.

3. Ignoring Inbox until requests pile up.

CHECK YOURSELF

Which surface should hold an approval request for an agent proposing a payment?

Answer: The Inbox, so a human explicitly approves or rejects before the action runs.

ENTERPRISE WORK & GEMINI · SERVICE 7

Connectors & BYO MCP

> *Reach Google Workspace, Microsoft 365, business apps and your own MCP servers.*

WHAT IT IS

Prebuilt connectors index or federate data from Workspace, Microsoft 365, Salesforce, SAP, ServiceNow, Jira and more, while bring your own MCP lets agents call internal systems through your servers.

WHY IT MATTERS

Agents are only as good as their context. Connectors deliver it securely and respect source permissions.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

Keys to every filing cabinet, each opening only the drawers you may see.

DEEP DIVE

Six things connectors gives you

01 Broad catalog

Connectors for productivity suites, CRM, ERP, ITSM and wikis.

02 Permission aware

Access control lists from the source system are honoured.

03 Ingest or federate

Index data for speed, or query live sources at request time.

04 Actions

Some connectors let agents create tickets, send mail or update records.

05 BYO MCP

Register your own MCP servers for proprietary systems.

06 Admin managed

IT controls which connectors exist and their scopes.

HOW IT WORKS · CONNECTORS & BYO MCP

From request to result

1

Connect

Admin authorises the source with least privilege scopes.

2

Sync

Content and ACLs index, or live federation is set up.

3

Retrieve

Queries fetch only what the user can access.

4

Act

Agents use connector or MCP actions with approval where needed.

WHERE IT SHINES

- Answering questions across SharePoint and Drive in one search.
- Creating Jira tickets from meeting notes.
- An MCP server exposing a mainframe inventory system.

Using it well in production

PRACTITIONER TIPS

1. Start read only; add write actions after review.
2. Monitor sync health and ACL freshness.
3. Document every BYO MCP server in Agent Registry.

⚠ PITFALLS TO AVOID

1. Connecting with a super admin account that sees everything.
2. Stale ACLs that expose revoked content.
3. Unreviewed write actions on systems of record.

CHECK YOURSELF

Why should connectors use least privilege service accounts?

Answer: So a compromised or misbehaving agent can reach only the data and actions it truly needs.

ENTERPRISE WORK & GEMINI · SERVICE 8

Antigravity

> *Agent first development environment and CLI for building software with Gemini.*

WHAT IT IS

Google's agent first IDE. Developers assign tasks to coding agents that plan, edit across files, run terminals and browsers, and report progress with artifacts such as plans, diffs and screenshots for review.

WHY IT MATTERS

It shifts developers from typing code to directing and reviewing agents, accelerating delivery of agents and apps.

LAYER 3 ·
ENTERPRISE WORK



THINK OF IT AS

A cockpit where you direct a crew of coding agents instead of typing every line.

CAPABILITIES · ANTIGRAVITY

DEEP DIVE

Six things Antigravity gives you

01 Agent manager

Run several agents in parallel on separate tasks.

02 Editor plus agents

A familiar editor with agents that act across the codebase.

03 Terminal and browser

Agents run commands and test web apps themselves.

04 Artifacts

Plans, task lists, diffs and screenshots make work reviewable.

05 Gemini powered

Uses Gemini models, with other models available.

06 Feedback loop

Comment on artifacts to steer agents mid task.

HOW IT WORKS · ANTIGRAVITY

From request to result

1

Assign

Describe the task, such as add an endpoint with tests.

2

Plan

Review the agent's implementation plan.

3

Execute

The agent edits, runs tests and checks the browser.

4

Review

Inspect diffs and artifacts, then accept or redirect.

WHERE IT SHINES

- Scaffolding an ADK agent with tests and deployment config.
- Parallel bug fixes across a backlog.
- UI changes verified by the agent in a browser.

FIELD NOTES · ANTIGRAVITY

Using it well in production

PRACTITIONER TIPS

1. Review plans before execution for risky changes.
2. Keep agents on branches and use pull requests.
3. Give agents test commands so they verify their own work.

⚠ PITFALLS TO AVOID

1. Accepting large diffs without reading them.
2. Letting agents run destructive commands in production shells.
3. Vague tasks that produce sprawling changes.

CHECK YOURSELF

What should a developer review before letting an Antigravity agent execute a risky change?

Answer: The implementation plan artifact, checking scope and approach before any code changes.

LAYER 3 RECAP

Three things to remember

1. The Gemini Enterprise app is one secure front door for chat, search and agents across all company data.
2. Every employee can build: Agent Designer and Workspace Studio for no code, Antigravity for developers.
3. Connectors and bring your own MCP servers decide how useful agents are; context is everything.

LAYER 3 QUIZ

Test yourself on Enterprise Work & Gemini

QUESTION 1

Where do employees find Google, in house and partner agents?

QUESTION 2

Which tool automates work inside Gmail, Docs and Sheets with no code?

QUESTION 3

What is Antigravity?

QUESTION 4

Name two non Google systems connectors can reach.

ANSWER KEY, FLIP TO READ

1. The Agent Gallery in the Gemini Enterprise app.
2. Workspace Studio.
3. Google's agent first development environment.
4. Microsoft 365, Salesforce, SAP, Jira and more.

LAYER FOUR · THE CONTEXT

Data, Knowledge & Memory

The context that makes agents useful. A unified data foundation, an active knowledge catalog, semantic understanding of work, developer data tooling, enterprise search, managed retrieval, vector search and long term memory.

BigQuery | Knowledge Catalog | Workspace Intelligence | Data Agent Kit | Agent Search | RAG Engine | Vector Search | Memory Bank

LAYER 4 MAP

The 8 services of Data, Knowledge & Memory

BigQuery
The company's central bank vault for data, with tellers who speak SQL and AI.

Knowledge Catalog
A map legend that tells agents what every road on the data map means.

Workspace Intelligence
An office manager who knows who works on what and where everything is.

Data Agent Kit
A toolbox that turns any coding assistant into a data specialist.

Agent Search
Google quality search, installed inside your own company.

RAG Engine
A research assistant who pulls the right pages before you answer.

Vector Search
A sorting machine that finds the closest meaning among billions in milliseconds.

Memory Bank
A thoughtful assistant's notebook about each client, kept across years.

DATA, KNOWLEDGE & MEMORY · SERVICE 1

BigQuery

> *The unified data foundation: warehouse, cross cloud lakehouse and built in AI functions.*

WHAT IT IS

Google's serverless data platform. It stores structured and unstructured data, runs SQL at petabyte scale, reads open Iceberg tables across clouds and calls Gemini directly from SQL with AI functions.

WHY IT MATTERS

Agents need trustworthy, current data. BigQuery puts analytics and AI on one governed foundation without moving data around.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

The company's central bank vault for data, with tellers who speak SQL and AI.

CAPABILITIES · BIGQUERY

DEEP DIVE

Six things BigQuery gives you

01 Serverless scale

Query terabytes to petabytes without managing clusters.

02 AI in SQL

Call Gemini from SQL to classify, extract and generate over rows.

03 Vector search

Store embeddings and run similarity search in BigQuery.

04 Lakehouse

Query Apache Iceberg tables and data in other clouds.

05 Data agents

Data engineering and data science agents help build pipelines and models.

06 MCP access

A managed MCP server lets agents query BigQuery safely.

HOW IT WORKS · BIGQUERY

From request to result

1

Land

Ingest or federate data into governed datasets.

2

Enrich

Use AI functions to tag, extract and summarise.

3

Expose

Publish curated views to agents via MCP or tools.

4

Answer

Agents query with SQL and explain results.

WHERE IT SHINES

- Sentiment tagging of millions of reviews in one SQL statement.
- An agent answering revenue questions from curated views.
- Joining CRM and support data for churn analysis.

FIELD NOTES · BIGQUERY

Using it well in production

PRACTITIONER TIPS

1. Expose curated views to agents, not raw tables.

2. Set cost controls such as maximum bytes billed for agent queries.

3. Document columns; agents read descriptions.

⚠ PITFALLS TO AVOID

1. Letting agents scan full raw tables on every question.

2. Undocumented columns with cryptic names.

3. Granting agents write access to production datasets.

CHECK YOURSELF

Why give agents curated views instead of raw tables?

Answer: Views limit scope, cost and sensitive columns, and present clean business logic the agent can rely on.

DATA, KNOWLEDGE & MEMORY · SERVICE 2

Knowledge Catalog

> *Formerly Dataplex Universal Catalog: metadata and business meaning for agents.*

WHAT IT IS

The evolution of Dataplex Universal Catalog into an active context engine. It curates metadata, lineage, quality, glossaries and relationships into a context graph that agents use to find and interpret data.

WHY IT MATTERS

Agents misread data without business context. The catalog tells them what data exists, what it means and whether to trust it.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

A map legend that tells agents what every road on the data map means.

CAPABILITIES · KNOWLEDGE CATALOG

DEEP DIVE

Six things Knowledge Catalog gives you

01 Discovery

Automatically finds and profiles data across lakes and warehouses.

02 Business glossary

Defines terms like active customer once for everyone.

03 Lineage

Shows where data came from and what depends on it.

04 Quality

Data quality scans flag broken or stale assets.

05 Context graph

Links data, meaning and rules into agent ready context.

06 Compatibility

Existing Dataplex APIs and configurations keep working.

HOW IT WORKS · KNOWLEDGE CATALOG

From request to result

1
Catalog
Discovery scans register assets and metadata.

2
Enrich
Stewards add glossary terms, owners and rules.

3
Serve
Agents query the catalog to pick trusted assets.

4
Trust
Quality and lineage signals guide answers.

WHERE IT SHINES

- A data agent picking the certified revenue table over a stale copy.
- Impact analysis before changing a pipeline.
- Consistent metric definitions across agents.

FIELD NOTES · KNOWLEDGE CATALOG

Using it well in production

PRACTITIONER TIPS

1. Start glossaries with the twenty most argued about metrics.
2. Assign owners to critical data assets.
3. Surface quality scores to agents and humans alike.

⚠ PITFALLS TO AVOID

1. Assuming automatic metadata replaces stewardship.
2. Multiple competing definitions of key terms.
3. Ignoring lineage until an incident.

CHECK YOURSELF

How does Knowledge Catalog reduce wrong agent answers?

Answer: It tells agents which assets are trusted and what business terms mean, so they query the right data correctly.

DATA, KNOWLEDGE & MEMORY · SERVICE 3

Workspace Intelligence

> *Semantic layer over mail, chats, files, people and active projects.*

WHAT IT IS

A semantic understanding of work inside Google Workspace. It relates emails, chats, documents, people and projects so Gemini and agents grasp context such as who owns a topic and what is currently active.

WHY IT MATTERS

Answers improve when agents understand relationships, not just keywords, for example the latest version of a plan and who decided it.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

An office manager who knows who works on what and where everything is.

CAPABILITIES · WORKSPACE INTELLIGENCE

DEEP DIVE

Six things Workspace Intelligence gives you

01 Entity understanding

Recognises people, projects, documents and threads.

02 Relationships

Connects who wrote what, who decided and what relates.

03 Recency

Knows which projects and files are currently active.

04 Permission aware

Uses only content each user can access.

05 Personal context

Tailors answers to your role and work.

06 Powers Gemini

Improves Gemini in Gmail, Docs, Chat and agents.

HOW IT WORKS · WORKSPACE INTELLIGENCE

From request to result

1
Observe
Workspace activity builds a semantic picture of work.

2
Relate
Entities and links form a graph of context.

3
Query
A user or agent asks a question in natural language.

4
Answer
Gemini retrieves the right, permitted context.

WHERE IT SHINES

- Asking what was decided about the Q3 launch and by whom.
- Finding the expert on a topic inside the company.
- Agents drafting updates from active project threads.

Using it well in production

PRACTITIONER TIPS

1. Keep files in shared drives with clear names to improve signals.

2. Teach staff to cite sources in Gemini answers.

3. Review data governance settings with admins.

⚠ PITFALLS TO AVOID

1. Expecting perfect answers from chaotic, unshared files.

2. Assuming it overrides sharing permissions; it does not.

3. Ignoring regional data residency requirements.

CHECK YOURSELF

Does Workspace Intelligence show users documents they cannot normally open?

Answer: No. It is permission aware and only uses content each user is allowed to access.

DATA, KNOWLEDGE & MEMORY · SERVICE 4

Data Agent Kit

> *Skills, MCP tools and IDE extensions for data engineering and data science agents.*

WHAT IT IS

A preview kit of skills, MCP tools and extensions that bring Google Cloud data capabilities into developer tools such as VS Code, Gemini CLI, Codex and Claude Code, grounded in Knowledge Catalog context.

WHY IT MATTERS

Data teams work in their IDE. The kit lets agents there build pipelines, notebooks and models with governed access to cloud data.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

A toolbox that turns any coding assistant into a data specialist.

CAPABILITIES · DATA AGENT KIT

DEEP DIVE

Six things the Data Agent Kit gives you

01 IDE extensions

Works inside VS Code and terminal based coding agents.

02 Data skills

Packaged know how for pipelines, SQL and analysis.

03 MCP tools

Agents query BigQuery, AlloyDB and Cloud SQL through MCP.

04 Catalog grounded

Uses Knowledge Catalog for definitions and lineage.

05 Agent choice

Works with Gemini CLI and other coding agents.

06 Data agents

Complements the Data Engineering and Data Science agents.

HOW IT WORKS · DATA AGENT KIT

From request to result

1

Install

Add the kit's extensions and MCP servers to your tools.

2

Ask

Request a pipeline, query or notebook in natural language.

3

Build

The agent writes code using governed context.

4

Ship

Review, test and deploy through normal workflows.

WHERE IT SHINES

- Generating a dbt style transformation with tests.
- Profiling a new dataset before modelling.
- Writing feature engineering SQL from a description.

FIELD NOTES · DATA AGENT KIT

Using it well in production

PRACTITIONER TIPS

1. Point agents at dev projects first, not production.
2. Keep generated SQL under code review.
3. Use catalog context so agents pick certified tables.

⚠ PITFALLS TO AVOID

1. Running agent written pipelines straight in production.
2. Granting broad data access to IDE agents.
3. Skipping tests on generated transformations.

CHECK YOURSELF

Which environments does the Data Agent Kit target?

Answer: Developer tools such as VS Code, Gemini CLI, Codex and Claude Code, via extensions, skills and MCP tools.

DATA, KNOWLEDGE & MEMORY · SERVICE 5

Agent Search

> *Formerly Vertex AI Search: enterprise search and grounding on your content.*

WHAT IT IS

A managed search and retrieval service. It ingests websites, documents and structured data, applies Google ranking, and powers search apps and grounding for agents with citations.

WHY IT MATTERS

Building search quality yourself is hard. Agent Search gives high quality retrieval in days instead of months.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

Google quality search, installed inside your own company.

CAPABILITIES · AGENT SEARCH

DEEP DIVE

Six things Agent Search gives you

01 Data stores

Index websites, Cloud Storage files, BigQuery tables and connectors.

02 Google ranking

Semantic and keyword ranking tuned from Google Search research.

03 Answers

Generative answers with citations over your content.

04 Grounding

Use as a grounding tool for Gemini and ADK agents.

05 Filters and boosts

Control results with metadata filters and boosting.

06 Blended search

Search several data stores in one query.

HOW IT WORKS · AGENT SEARCH

From request to result

1

Create store

Point at sources; ingestion parses and indexes.

2

Configure

Set filters, boosts and answer settings.

3

Query

Apps or agents send search requests.

4

Ground

Results and citations feed the model's answer.

WHERE IT SHINES

- Intranet search across policies and product docs.
- Customer self service search on a support site.
- Grounding a sales agent on product specs.

Using it well in production

PRACTITIONER TIPS

1. Clean source documents; headings and titles improve ranking.
2. Add metadata for filtering by region, product or date.
3. Measure search quality with real queries.

⚠ PITFALLS TO AVOID

1. Indexing duplicates and outdated versions.
2. No metadata, so filtering is impossible.
3. Skipping relevance evaluation.

CHECK YOURSELF

When would you choose Agent Search over building your own RAG pipeline?

Answer: When you want high quality managed retrieval fast and do not need to control chunking and ranking internals.

DATA, KNOWLEDGE & MEMORY · SERVICE 6

RAG Engine

> *Managed retrieval that connects private data to models for accurate answers.*

WHAT IT IS

A managed retrieval augmented generation framework. It handles ingestion, parsing, chunking, embedding, indexing and retrieval into a corpus, with a choice of vector databases, then feeds results to models.

WHY IT MATTERS

RAG Engine gives more control than turnkey search while avoiding the plumbing of a fully custom pipeline.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

A research assistant who pulls the right pages before you answer.

CAPABILITIES · RAG ENGINE

DEEP DIVE

Six things RAG Engine gives you

01 Corpus management

Create corpora and import files from Storage, Drive and more.

02 Chunking control

Configure chunk size and overlap for your documents.

03 Embedding choice

Select the embedding model used for vectors.

04 Vector backends

Managed database, Vector Search or third party stores.

05 Retrieval tuning

Set top K, filters and reranking.

06 Grounding tool

Attach the corpus as a retrieval tool to Gemini.

HOW IT WORKS · RAG ENGINE

From request to result

1

Import

Load and parse files into a corpus.

2

Chunk and embed

Split text and create vectors.

3

Retrieve

Find the most relevant chunks for a query.

4

Generate

Gemini answers grounded on retrieved chunks.

WHERE IT SHINES

- A legal assistant over thousands of contracts.
- Engineering Q and A over design documents.
- Per customer corpora for multi tenant apps.

FIELD NOTES · RAG ENGINE

Using it well in production

PRACTITIONER TIPS

1. Tune chunk size on your documents; there is no universal best.

2. Store source metadata for citations.

3. Evaluate retrieval separately from generation.

⚠ PITFALLS TO AVOID

1. Chunks so small they lose meaning.

2. Never refreshing the corpus as documents change.

3. Blaming the model for retrieval failures.

CHECK YOURSELF

Why evaluate retrieval separately from generation in RAG?

Answer: If the right chunks are not retrieved, no model can answer correctly; isolating retrieval shows where to fix.

Vector Search

> *AI native engine to store and search embeddings at scale.*

WHAT IT IS

A fast approximate nearest neighbour service built on Google's ScaNN research. It stores embeddings and returns the most similar items with low latency at very large scale, with filtering and hybrid search.

WHY IT MATTERS

Recommendations, semantic search and RAG all depend on fast similarity search. Vector Search delivers it as a managed service.



THINK OF IT AS

A sorting machine that finds the closest meaning among billions in milliseconds.

CAPABILITIES · VECTOR SEARCH

DEEP DIVE

Six things Vector Search gives you

01 Billion scale

Searches very large vector collections with low latency.

02 ScaNN

Built on Google's research for fast approximate search.

03 Hybrid search

Combine dense semantic vectors with sparse keyword signals.

04 Filtering

Restrict results by attributes such as tenant or category.

05 Streaming updates

Add or update vectors without full rebuilds.

06 Autoscaling

Endpoints scale with query load.

HOW IT WORKS · VECTOR SEARCH

From request to result

1
Embed
Create vectors for items with an embedding model.

2
Index
Build an index and deploy it to an endpoint.

3
Query
Embed the query and search for nearest neighbours.

4
Use
Return items to an app, recommender or RAG step.

WHERE IT SHINES

- Product recommendations from similar item embeddings.
- Semantic search over millions of support articles.
- Deduplicating near identical records.

FIELD NOTES · VECTOR SEARCH

Using it well in production

PRACTITIONER TIPS

1. Use filters for multi tenant isolation.
2. Benchmark recall versus latency settings.
3. Combine with keyword signals for codes and names.

⚠ PITFALLS TO AVOID

1. Pure semantic search failing on part numbers.
2. No tenant filter in shared indexes.
3. Oversized endpoints for small workloads.

CHECK YOURSELF

Why add hybrid keyword signals to vector search?

Answer: Exact terms like product codes and names are matched poorly by meaning alone; keywords catch them.

DATA, KNOWLEDGE & MEMORY · SERVICE 8

Memory Bank

> *Long term agent memory across sessions, with Memory Profiles.*

WHAT IT IS

Managed long term memory. It extracts meaningful facts from conversations, consolidates them, stores them per user scope and retrieves relevant memories in future sessions, with revisions and access control.

WHY IT MATTERS

Users hate repeating themselves. Memory Bank lets agents remember preferences and context across sessions, safely and selectively.

LAYER 4 · DATA AND MEMORY



THINK OF IT AS

A thoughtful assistant's notebook about each client, kept across years.

CAPABILITIES · MEMORY BANK

DEEP DIVE

Six things Memory Bank gives you

01 Extraction

Gemini pulls durable facts from conversation events.

02 Consolidation

New facts merge with or update existing memories.

03 Scopes

Memories are stored per user or other defined scope.

04 Similarity retrieval

Fetch memories relevant to the current request.

05 Memory Profiles

Shape what kinds of facts are remembered.

06 Revisions and IAM

Inspect memory history and control access with IAM Conditions.

HOW IT WORKS · MEMORY BANK

From request to result

1
Converse
Session events are recorded as the user chats.

2
Generate
Memory Bank extracts and consolidates facts.

3
Retrieve
Next session, relevant memories load into context.

4
Personalise
The agent answers with remembered preferences.

WHERE IT SHINES

- Remembering a customer's preferred delivery window.
- A tutor agent tracking which topics a learner mastered.
- An assistant recalling a manager's report format.

FIELD NOTES · MEMORY BANK

Using it well in production

PRACTITIONER TIPS

1. Define what should and should not be remembered.

2. Let users view and delete their memories.

3. Set time to live for transient facts.

⚠ PITFALLS TO AVOID

1. Remembering sensitive data with no retention policy.

2. Loading every memory into every prompt.

3. Memories shared across users through wrong scopes.

CHECK YOURSELF

Why should users be able to view and delete what an agent remembers?

Answer: It builds trust, supports privacy rights and lets users correct wrong memories.

LAYER 4 RECAP

Three things to remember

1. BigQuery and Knowledge Catalog give agents governed data plus the business meaning behind it.
2. Agent Search, RAG Engine and Vector Search are three levels of retrieval, from turnkey to fully custom.
3. Memory Bank turns one off conversations into lasting, personalised relationships across sessions.

LAYER 4 QUIZ

Test yourself on Data, Knowledge & Memory

QUESTION 1

What was Knowledge Catalog formerly called?

QUESTION 2

Which service gives turnkey enterprise search and grounding?

QUESTION 3

Where do long term user preferences across sessions live?

QUESTION 4

Which kit brings data agents into VS Code and CLIs?

ANSWER KEY, FLIP TO READ

1. Dataplex Universal Catalog.
2. Agent Search, formerly Vertex AI Search.
3. Memory Bank.
4. Data Agent Kit.

LAYER FIVE · THE TRUST

Identity, Security & Governance

The controls that let enterprises trust agents. Every agent gets an identity, every tool and agent is registered, every call passes a policy gateway, and security posture, threats and data perimeters are watched end to end.

Agent Identity | Agent Registry | Agent Gateway | Model Armor | Security Command Center | Wiz | Google Security Operations | IAM & VPC Service Controls

LAYER 5 MAP

The 8 services of Identity, Security & Governance

Agent Identity

A staff ID badge for each digital worker, scanned at every door.

Agent Registry

The company register of who works here and which tools they may use.

Agent Gateway

Airport security that every agent and tool call must pass through.

Model Armor

A bodyguard screening every message in and out of the model.

Security Command Center

The security operations wall of screens for your whole cloud estate.

Wiz

An X ray that sees every AI component across all your clouds.

Google Security Operations

A security team that never sleeps and reads every alert.

IAM & VPC Service Controls

Locks, fences and your own keys around the whole estate.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 1

Agent Identity

> *A unique cryptographic identity for every agent: assign, authorize, audit.*

WHAT IT IS

A dedicated identity for each agent, distinct from human users and shared service accounts. Agents authenticate with their own credentials, receive scoped permissions and leave auditable trails of every action.

WHY IT MATTERS

When agents act, you must know which agent did what. Unique identity makes least privilege and accountability possible.

LAYER 5 · SECURITY



THINK OF IT AS

A staff ID badge for each digital worker, scanned at every door.

CAPABILITIES · AGENT IDENTITY

DEEP DIVE

Six things Agent Identity gives you

01 Unique per agent

Each agent gets its own identity, not a shared key.

02 Cryptographic proof

Strong credentials prove an agent is who it claims.

03 Assign

Bind identities to agents at deployment.

04 Authorize

Grant narrow IAM roles to each agent identity.

05 Audit

Every action is logged against the acting agent.

06 On behalf of users

Supports acting with delegated user authority where configured.

HOW IT WORKS · AGENT IDENTITY

From request to result

1

Deploy

An agent is created and receives its identity.

2

Grant

Admins attach least privilege roles.

3

Act

The agent calls tools and services with its credentials.

4

Trace

Audit logs attribute each call to the agent.

WHERE IT SHINES

- Proving which agent changed a customer record.
- Revoking one misbehaving agent without touching others.
- Different permissions for read only and write capable agents.

FIELD NOTES · AGENT IDENTITY

Using it well in production

PRACTITIONER TIPS

1. One identity per agent and per environment.

2. Review agent permissions quarterly like employee access.

3. Alert on unusual agent behaviour by identity.

⚠ PITFALLS TO AVOID

1. Many agents sharing one powerful service account.

2. Long lived broad credentials embedded in code.

3. No owner recorded for an agent identity.

CHECK YOURSELF

Why not let all agents share one service account?

Answer: You lose attribution and least privilege; one compromised agent would have every agent's access.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 2

Agent Registry

> *Central catalog of approved agents, tools and MCP servers: discover, track, manage.*

WHAT IT IS

A governed inventory of agents, tools and MCP servers across the organisation. It records owners, versions, capabilities and approval status so teams can discover reusable components and admins can control them.

WHY IT MATTERS

Agent sprawl creates shadow AI. A registry gives visibility and a single place to approve, find and retire components.

LAYER 5 · SECURITY



THINK OF IT AS

The company register of who works here and which tools they may use.

DEEP DIVE

Six things Agent Registry gives you

01 Inventory

Every agent, tool and MCP server in one catalog.

02 Metadata

Owners, versions, descriptions and data access.

03 Approval status

Mark components approved, pending or retired.

04 Discovery

Teams find and reuse existing agents and tools.

05 Policy input

Gateway policies reference registry entries.

06 Lifecycle

Track deprecations and replacements.

HOW IT WORKS · AGENT REGISTRY

From request to result

1

Register

Teams submit agents and MCP servers.

2

Review

Security and owners approve.

3

Discover

Builders reuse approved components.

4

Retire

Old versions are deprecated and blocked.

WHERE IT SHINES

Finding an approved CRM MCP server instead of building another.

Auditors listing all agents with access to personal data.

Blocking an unapproved tool at the gateway.

FIELD NOTES · AGENT REGISTRY

Using it well in production

PRACTITIONER TIPS

1. Make registration part of the deployment pipeline.

2. Require an owner and data classification per entry.

3. Publish a reuse first policy for tools.

⚠ PITFALLS TO AVOID

1. Registries filled manually and never updated.

2. No link between registry status and enforcement.

3. Duplicate servers for the same system.

CHECK YOURSELF

How does Agent Registry reduce shadow AI?

Answer: It inventories every agent and tool, and unregistered components can be blocked by gateway policy.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 3

Agent Gateway

> *One policy point for agent to agent and agent to tool traffic over MCP and A2A.*

WHAT IT IS

The networking and policy enforcement component of Agent Platform. All agent traffic over MCP and A2A, including across clouds, passes through it so access policies, inspection and logging apply consistently.

WHY IT MATTERS

Without a chokepoint, each agent enforces its own rules. A gateway centralises policy, Model Armor screening and audit.

LAYER 5 · SECURITY



THINK OF IT AS

Airport security that every agent and tool call must pass through.

CAPABILITIES · AGENT GATEWAY

DEEP DIVE

Six things Agent Gateway gives you

01 Single chokepoint

Agent to agent and agent to tool calls route through it.

02 Protocol aware

Understands MCP and A2A, not just raw HTTP.

03 Policy enforcement

Allow or deny by agent identity, tool and action.

04 Model Armor

Integrates screening for injection and data leakage.

05 Cross cloud

Governs agent traffic beyond Google Cloud.

06 Logging

Records every call for audit and investigation.

HOW IT WORKS · AGENT GATEWAY

From request to result

1

Call

An agent calls a tool or agent through the gateway.

2

Check

Identity, registry status and policy are evaluated.

3

Screen

Model Armor inspects prompts and responses.

4

Forward

Allowed calls pass; all are logged.

WHERE IT SHINES

- Blocking agents from calling unapproved MCP servers.
- Restricting a finance tool to finance agents only.
- Investigating an incident from gateway logs.

FIELD NOTES · AGENT GATEWAY

Using it well in production

PRACTITIONER TIPS

1. Default deny, then allow specific agent and tool pairs.
2. Pair gateway policy with registry status.
3. Check regional availability for your deployment.

⚠ PITFALLS TO AVOID

1. Bypass paths where agents call tools directly.
2. Overly broad allow rules.
3. Not monitoring blocked call patterns.

CHECK YOURSELF

What is the value of a single chokepoint for agent traffic?

Answer: Policies, screening and logging apply uniformly, so no agent can quietly skip controls.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 4

Model Armor

> *Runtime protection against prompt injection, tool poisoning and data leaks.*

WHAT IT IS

A runtime screening service that inspects prompts and responses for prompt injection, jailbreaks, malicious URLs, sensitive data and harmful content, applying configurable templates across models and agents.

WHY IT MATTERS

Agents read untrusted content and hold sensitive data. Model Armor blocks attacks and leaks independently of the model.

LAYER 5 · SECURITY



THINK OF IT AS

A bodyguard screening every message in and out of the model.

DEEP DIVE

Six things Model Armor gives you

01 Injection detection

Flags prompt injection and jailbreak attempts.

02 Sensitive data

Detects and can redact personal and confidential data.

03 Malicious URLs

Screens links in prompts and outputs.

04 Content safety

Filters harmful categories by configurable thresholds.

05 Model agnostic

Works across Gemini, partner and open models.

06 Templates

Reusable policies with organisation floor settings.

HOW IT WORKS · MODEL ARMOR

From request to result

1
Intercept
A prompt or tool result arrives.

2
Scan
Model Armor checks against the template.

3
Decide
Allow, redact or block based on findings.

4
Log
Findings flow to security monitoring.

WHERE IT SHINES

- Stopping a web page from instructing an agent to exfiltrate data.
- Redacting card numbers before prompts reach a model.
- Screening tool outputs for poisoned instructions.

Using it well in production

PRACTITIONER TIPS

1. Screen both inputs and outputs, including tool results.

2. Set organisation floor settings so no team goes below baseline.

3. Tune thresholds with real traffic samples.

⚠ PITFALLS TO AVOID

1. Screening only user prompts, not retrieved content.

2. Thresholds so strict users abandon the agent.

3. No alerting on repeated blocked attacks.

CHECK YOURSELF

Why screen tool results, not just user prompts?

Answer: Injected instructions often hide in retrieved pages or tool outputs the agent reads, not in the user's own message.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 5

Security Command Center

> *Agent security dashboard for asset discovery, threat detection and vulnerabilities.*

WHAT IT IS

Google Cloud's security and risk platform. It discovers assets including AI workloads and agents, detects threats, finds misconfigurations and vulnerabilities, and prioritises risks in one dashboard.

WHY IT MATTERS

You cannot protect agents you cannot see. SCC brings AI assets into the same posture and threat view as the rest of cloud.

LAYER 5 · SECURITY



THINK OF IT AS

The security operations wall of screens for your whole cloud estate.

DEEP DIVE

Six things Security Command Center gives you

01 Asset discovery

Inventories models, endpoints, agents and data.

02 AI protection

Highlights risks specific to AI workloads.

03 Threat detection

Detects suspicious activity in projects.

04 Vulnerabilities

Scans for exploitable weaknesses.

05 Misconfigurations

Flags public endpoints and weak IAM.

06 Prioritisation

Ranks findings by exposure and impact.

HOW IT WORKS · SECURITY COMMAND CENTER

From request to result

1

Discover

SCC inventories cloud and AI assets.

2

Assess

It evaluates posture and vulnerabilities.

3

Detect

It raises threats and anomalies.

4

Remediate

Teams fix issues, often with guided steps.

WHERE IT SHINES

- Finding a publicly exposed model endpoint.
- Detecting unusual data access by an agent.
- Reporting AI security posture to the board.

Using it well in production

PRACTITIONER TIPS

1. Turn on AI asset discovery early.
2. Route high findings to on call owners.
3. Track mean time to remediate.

⚠ PITFALLS TO AVOID

1. Alert fatigue from untriaged findings.
2. Ignoring development projects.
3. No owner mapping for assets.

CHECK YOURSELF

What does Security Command Center add for AI agents specifically?

Answer: Discovery of AI assets and AI specific risk findings alongside normal cloud posture and threat detection.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 6

Wiz

> *AI security posture across clouds and agent studios, with an AI bill of materials.*

WHAT IT IS

Google Cloud's cloud native application protection platform. It maps cloud resources, code and AI components across providers, builds an AI bill of materials and finds toxic combinations of risk.

WHY IT MATTERS

Agents span clouds and SaaS. Wiz gives a single, cross cloud view of AI risk, including where models and data are exposed.

LAYER 5 · SECURITY



THINK OF IT AS

An X ray that sees every AI component across all your clouds.

CAPABILITIES · WIZ

DEEP DIVE

Six things Wiz gives you

01 Multi cloud

Covers Google Cloud, AWS, Azure and more.

02 AI BOM

Inventory of models, SDKs, pipelines and data in use.

03 Security graph

Links resources to reveal attack paths.

04 Toxic combinations

Finds risk where exposure meets sensitive data.

05 Agentless

Scans without installing agents on workloads.

06 AI app protection

Extends posture management to AI applications.

HOW IT WORKS · WIZ

From request to result

1
Connect
Link cloud accounts and code repositories.

2
Map
Wiz builds a graph and AI bill of materials.

3
Prioritise
Attack paths rank critical issues.

4
Fix
Teams remediate with owners and tickets.

WHERE IT SHINES

- Finding shadow AI SDKs in production code.
- Spotting a public bucket feeding a training pipeline.
- Cross cloud reporting on AI usage.

FIELD NOTES · WIZ

Using it well in production

PRACTITIONER TIPS

1. Use the AI BOM to drive governance reviews.
2. Focus on attack paths, not raw finding counts.
3. Integrate findings into developer workflows.

⚠ PITFALLS TO AVOID

1. Treating AI risk separately from cloud risk.
2. Ignoring other clouds where agents run.
3. Exporting findings nobody owns.

CHECK YOURSELF

What is an AI bill of materials used for?

Answer: To know exactly which models, libraries, data and services make up your AI systems, so risk can be managed.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 7

Google Security Operations

Agents for alert triage, threat hunting and detection engineering.

WHAT IT IS

Google's security operations platform combining SIEM, SOAR and threat intelligence, with Gemini powered agents that triage alerts, investigate incidents, hunt threats and help write detections.

WHY IT MATTERS

Security teams are overloaded. Agents handle repetitive triage so analysts focus on real attacks, including attacks on agents.

LAYER 5 · SECURITY



THINK OF IT AS

A security team that never sleeps and reads every alert.

CAPABILITIES · GOOGLE SECURITY OPERATIONS

DEEP DIVE

Six things Google SecOps gives you

01 SIEM scale

Ingests and searches security telemetry at large scale.

02 Triage agent

Investigates alerts and summarises verdicts with evidence.

03 Threat hunting

Natural language hunts across logs.

04 Detection engineering

Helps write and tune detection rules.

05 Threat intelligence

Mandiant and Google intelligence enrich findings.

06 SOAR playbooks

Automates response steps with approvals.

HOW IT WORKS · GOOGLE SECURITY OPERATIONS

From request to result

1

Ingest

Logs from cloud, endpoints and agents flow in.

2

Triage

Agents investigate alerts and gather evidence.

3

Decide

Analysts review verdicts and escalate.

4

Respond

Playbooks contain and remediate.

WHERE IT SHINES

- Auto triaging phishing reports.
- Hunting for abuse of agent credentials.
- Drafting detections for new attack techniques.

Using it well in production

PRACTITIONER TIPS

1. Feed agent and gateway logs into SecOps.
2. Keep human approval for containment actions.
3. Measure analyst time saved and accuracy.

⚠ PITFALLS TO AVOID

1. Fully automated response without review.
2. Missing logs from agent infrastructure.
3. Trusting verdicts without evidence.

CHECK YOURSELF

Why feed Agent Gateway logs into Google SecOps?

Answer: So attacks on or by agents can be detected, investigated and responded to like any other threat.

IDENTITY, SECURITY & GOVERNANCE · SERVICE 8

IAM & VPC Service Controls

> *Access control, service perimeters and customer managed encryption keys.*

WHAT IT IS

The foundational controls. IAM decides who and which agents can do what, VPC Service Controls draw perimeters that stop data exfiltration, and CMEK lets you own encryption keys for data at rest.

WHY IT MATTERS

Every higher layer relies on these basics. Misconfigured IAM or missing perimeters undo all other AI security.

LAYER 5 · SECURITY



THINK OF IT AS

Locks, fences and your own keys around the whole estate.

CAPABILITIES · IAM & VPC SERVICE CONTROLS

DEEP DIVE

Six things IAM and VPC SC gives you

01 IAM roles

Fine grained roles for users, groups and agent identities.

02 Conditions

Context aware rules such as time, resource or attribute.

03 Service perimeters

Block data leaving trusted boundaries.

04 CMEK

Customer managed keys encrypt stored data.

05 Audit logs

Who did what, where and when.

06 Org policies

Guardrails applied across all projects.

HOW IT WORKS · IAM & VPC SERVICE CONTROLS

From request to result

1

Design

Plan roles, groups and perimeters.

2

Apply

Grant least privilege and enable perimeters.

3

Encrypt

Configure CMEK for sensitive services.

4

Review

Audit access and policy violations regularly.

WHERE IT SHINES

- Preventing an agent from copying data to an outside project.
- Meeting regulator demands for customer owned keys.
- Restricting Memory Bank access with IAM Conditions.

Using it well in production

PRACTITIONER TIPS

1. Use groups, not individual grants.
2. Test perimeters in dry run mode first.
3. Rotate and monitor keys.

⚠ PITFALLS TO AVOID

1. Primitive owner or editor roles on agent identities.
2. Perimeters with broad bypass rules.
3. Losing access to keys and therefore data.

CHECK YOURSELF

What do VPC Service Controls prevent?

Answer: Data exfiltration across a defined perimeter, even by identities with valid credentials.

LAYER 5 RECAP

Three things to remember

1. Treat agents as first class identities: assign, authorize and audit them like employees, not shared keys.
2. Registry plus Gateway plus Model Armor form the control plane for all agent to agent and agent to tool traffic.
3. Wiz, Security Command Center and SecOps extend cloud security to AI assets, models and agent behaviour.

LAYER 5 QUIZ

Test yourself on Identity, Security & Governance

QUESTION 1

Which service gives every agent a unique cryptographic identity?

QUESTION 2

Where is one policy point for MCP and A2A traffic?

QUESTION 3

What does Model Armor protect against?

QUESTION 4

What is an AI bill of materials?

ANSWER KEY, FLIP TO READ

1. Agent Identity.
2. Agent Gateway.
3. Prompt injection, tool poisoning and data leaks.
4. An inventory of models, data and AI components in use.

LAYER SIX · THE OPERATIONS

Production & Deployment

Everything needed to run agents reliably. Tracing, evaluation on live traffic, simulation before launch, automatic optimisation, monitoring, compute choices, sovereign deployment, developer CLIs and a full CI/CD path.

Agent Observability | Agent Evaluation | Agent Simulation | Agent Optimizer | Cloud Monitoring & Logging | Cloud Run & GKE | Cloud TPUs & GPUs | Google Distributed Cloud | Agents CLI | Cloud Build & Cloud Deploy

LAYER 6 MAP

The 10 services of Production & Deployment

Agent Observability
A flight recorder for every agent decision.

Agent Evaluation
A quality inspector who samples every production shift.

Agent Simulation
A flight simulator where agents crash safely before carrying passengers.

Agent Optimizer
A coach reviewing match footage and rewriting the playbook.

Cloud Monitoring & Logging
The building's alarm panel, CCTV and visitor log in one.

Cloud Run & GKE
Choosing between a serviced apartment and owning the whole building.

Cloud TPUs & GPUs
The engines in the factory, purpose built for AI work.

Google Distributed Cloud
A branch office of the cloud built inside your own walls.

Agents CLI
A command line foreman that assembles agents to specification.

Cloud Build & Cloud Deploy
An assembly line with quality checks at every station.

PRODUCTION & DEPLOYMENT · SERVICE 1

Agent Observability

> *Unified trace viewer for agent reasoning, tool calls and latency.*

WHAT IT IS

Tracing and dashboards that capture each step an agent takes: model calls, reasoning, tool invocations, sub agent handoffs, tokens and latency, built on OpenTelemetry and Cloud Trace.

WHY IT MATTERS

Agents fail in surprising ways. Traces show exactly where and why, turning debugging from guesswork into inspection.

LAYER 6 · PRODUCTION



THINK OF IT AS

A flight recorder for every agent decision.

CAPABILITIES · AGENT OBSERVABILITY

DEEP DIVE

Six things Agent Observability gives you

01 Step traces

Spans for each model call, tool call and handoff.

02 Reasoning view

See intermediate thoughts and decisions where available.

03 Latency breakdown

Find which step makes a request slow.

04 Token usage

Track cost drivers per step and per agent.

05 OpenTelemetry

Standard instrumentation works across frameworks.

06 Session linking

Connect traces to sessions and users.

HOW IT WORKS · AGENT OBSERVABILITY

From request to result

1

Instrument

Enable tracing in ADK or OpenTelemetry.

2

Collect

Spans flow to Cloud Trace.

3

Inspect

Open the trace viewer for a failing request.

4

Fix

Change prompts, tools or routing based on evidence.

WHERE IT SHINES

- Finding a slow tool call adding five seconds.
- Seeing why an agent picked the wrong sub agent.
- Attributing token spend to a runaway loop.

Using it well in production

PRACTITIONER TIPS

1. Enable tracing from the first prototype.
2. Redact sensitive data in spans.
3. Link traces to eval failures.

⚠ PITFALLS TO AVOID

1. Logging full prompts with personal data.
2. Only aggregated metrics, no per request traces.
3. Sampling so low that failures are missed.

CHECK YOURSELF

A user reports a slow answer. What do you open first?

Answer: The trace for that request, to see which model or tool step consumed the time.

PRODUCTION & DEPLOYMENT · SERVICE 2

Agent Evaluation

> *Multi turn autoraters score quality on live production traffic.*

WHAT IT IS

Evaluation for agents, not just single prompts. It scores multi turn conversations and tool trajectories with autoraters and rubrics, both offline on test sets and continuously on sampled live traffic.

WHY IT MATTERS

Agents drift as data, tools and users change. Continuous evaluation catches regressions before customers do.

LAYER 6 · PRODUCTION



THINK OF IT AS

A quality inspector who samples every production shift.

CAPABILITIES · AGENT EVALUATION

DEEP DIVE

Six things Agent Evaluation gives you

01 Multi turn

Scores whole conversations, not just one reply.

02 Trajectory checks

Did the agent call the right tools in the right order?

03 Live sampling

Evaluate a share of real production traffic.

04 Rubrics

Business specific criteria such as policy compliance.

05 Trends

Quality dashboards over time and versions.

06 Gate releases

Block deploys that fall below thresholds.

HOW IT WORKS · AGENT EVALUATION

From request to result

1

Define

Write rubrics and trajectory expectations.

2

Sample

Pull test sets or live conversations.

3

Score

Autoraters grade and explain.

4

Act

Alert, roll back or improve on regressions.

WHERE IT SHINES

- Detecting a quality drop after a tool API changed.
- Comparing two agent versions in production.
- Compliance scoring of every hundredth conversation.

FIELD NOTES · AGENT EVALUATION

Using it well in production

PRACTITIONER TIPS

1. Mix offline gates with online monitoring.
2. Review autorater disagreements with humans.
3. Version rubrics with the agent.

⚠ PITFALLS TO AVOID

1. Only checking final answers, ignoring tool misuse.
2. Evaluating once at launch.
3. Rubrics nobody owns.

CHECK YOURSELF

Why evaluate tool trajectories, not just final answers?

Answer: An agent can reach a plausible answer through wrong or risky tool calls; trajectories reveal that.

PRODUCTION & DEPLOYMENT · SERVICE 3

Agent Simulation

> *Synthetic users and virtual tools stress test agents before launch.*

WHAT IT IS

A pre launch testing service that generates synthetic users with varied goals and personas, and virtual tools that mimic real systems, so agents face thousands of realistic scenarios safely.

WHY IT MATTERS

Real users find edge cases fast. Simulation finds them first, without touching production systems or customers.

LAYER 6 · PRODUCTION



THINK OF IT AS

A flight simulator where agents crash safely before carrying passengers.

CAPABILITIES · AGENT SIMULATION

DEEP DIVE

Six things Agent Simulation gives you

01 Synthetic users

Personas with goals, moods and tricky behaviour.

02 Virtual tools

Mocked systems return realistic responses and errors.

03 Scale

Run thousands of conversations in parallel.

04 Adversarial cases

Include injection attempts and abusive users.

05 Scored runs

Results feed Agent Evaluation.

06 Safe

No real data or systems are touched.

HOW IT WORKS · AGENT SIMULATION

From request to result

1

Design

Define personas, goals and tool behaviours.

2

Simulate

Run conversations at scale.

3

Score

Evaluate success, safety and efficiency.

4

Fix

Address failure clusters before launch.

WHERE IT SHINES

- Testing a refund agent with angry and confused customers.
- Seeing how agents handle a tool timing out.
- Red teaming against prompt injection.

Using it well in production

PRACTITIONER TIPS

1. Seed personas from real support transcripts.
2. Simulate tool errors, not just success.
3. Re run simulations on every major change.

⚠ PITFALLS TO AVOID

1. Only polite, happy path synthetic users.
2. Virtual tools that never fail.
3. Ignoring simulation results under deadline pressure.

CHECK YOURSELF

Why should virtual tools in simulation return errors and timeouts?

Answer: Real systems fail; agents must be proven to handle failures gracefully before launch.

PRODUCTION & DEPLOYMENT · SERVICE 4

Agent Optimizer

> *Clusters real failures and proposes better instructions.*

WHAT IT IS

A service that analyses failed or low scoring interactions, groups them into failure clusters and proposes improved agent instructions or configurations, which you test before adopting.

WHY IT MATTERS

Manually reading thousands of conversations is impossible. Optimizer turns failure data into concrete, testable fixes.

LAYER 6 · PRODUCTION



THINK OF IT AS

A coach reviewing match footage and rewriting the playbook.

CAPABILITIES · AGENT OPTIMIZER

DEEP DIVE

Six things Agent Optimizer gives you

01 Failure clustering

Groups similar failures into themes.

02 Root cause hints

Explains likely causes per cluster.

03 Instruction proposals

Suggests revised prompts and settings.

04 Eval backed

Proposals are tested against eval sets.

05 Human approval

Teams review before changes ship.

06 Continuous loop

Runs as new production data arrives.

HOW IT WORKS · AGENT OPTIMIZER

From request to result

1

Collect

Gather low scoring sessions from evaluation.

2

Cluster

Optimizer groups failures by pattern.

3

Propose

It drafts improved instructions.

4

Validate

Evals confirm improvement; then deploy.

WHERE IT SHINES

- Fixing an agent that misreads date formats from one region.
- Reducing unnecessary tool calls.
- Improving tone in complaint conversations.

FIELD NOTES · AGENT OPTIMIZER

Using it well in production

PRACTITIONER TIPS

1. Always validate proposals on held out evals.
2. Change one thing at a time to learn what worked.
3. Keep a changelog of instruction edits.

⚠ PITFALLS TO AVOID

1. Auto applying proposals without review.
2. Optimising for the eval set only.
3. Ignoring clusters that need tool fixes, not prompt fixes.

CHECK YOURSELF

Why validate an Optimizer proposal on held out evaluation data?

Answer: To confirm the change truly improves behaviour and does not just overfit the failures it was built from.

PRODUCTION & DEPLOYMENT · SERVICE 5

Cloud Monitoring & Logging

Cloud Trace monitors agent metrics, logs and audit trails.

WHAT IT IS

Google Cloud's operations suite: metrics, dashboards, alerts, centralised logs and audit trails covering agent infrastructure, model endpoints and usage, linked with traces.

WHY IT MATTERS

Reliability needs alerts before users complain, and compliance needs durable audit evidence. This suite provides both.

LAYER 6 · PRODUCTION



THINK OF IT AS

The building's alarm panel, CCTV and visitor log in one.

DEEP DIVE

Six things Monitoring and Logging gives you

01 Metrics

Latency, errors, tokens and throughput per agent.

02 Dashboards

Ready made and custom views for agent fleets.

03 Alerting

Notify on error spikes, latency or cost anomalies.

04 Central logs

Search and route logs from all services.

05 Audit logs

Admin and data access records for compliance.

06 SLOs

Define service level objectives and error budgets.

HOW IT WORKS · CLOUD MONITORING & LOGGING

From request to result

1

Emit

Agents and services send metrics and logs.

2

Watch

Dashboards and SLOs track health.

3

Alert

Policies notify on call engineers.

4

Investigate

Jump from alert to logs and traces.

WHERE IT SHINES

- Paging on call when agent error rate exceeds two percent.
- Monthly audit evidence of data access.
- Spotting a token cost spike overnight.

Using it well in production

PRACTITIONER TIPS

1. Define SLOs for latency and success rate.
2. Route logs to long term storage per retention rules.
3. Add cost alerts for model usage.

⚠ PITFALLS TO AVOID

1. Alerts without owners.
2. Logging secrets or personal data.
3. No retention policy for audit logs.

CHECK YOURSELF

What is an error budget used for?

Answer: It quantifies acceptable unreliability under an SLO, guiding when to ship features versus fix stability.

PRODUCTION & DEPLOYMENT · SERVICE 6

Cloud Run & GKE

> *Serverless containers or Kubernetes alongside managed Agent Runtime.*

WHAT IT IS

Two compute options for agents and tools. Cloud Run runs containers serverlessly with scale to zero; GKE gives full Kubernetes control, including sandboxed agent workloads and GPU or TPU nodes.

WHY IT MATTERS

Some agents need custom runtimes, networking or hardware. These options give control while staying in the platform.

LAYER 6 · PRODUCTION



THINK OF IT AS

Choosing between a serviced apartment and owning the whole building.

CAPABILITIES · CLOUD RUN & GKE

DEEP DIVE

Six things Cloud Run and GKE gives you

01 Cloud Run

Deploy a container; it scales automatically, even to zero.

02 GKE

Managed Kubernetes for complex, stateful or high scale systems.

03 Agent sandboxing

GKE offers sandboxed environments for agent code.

04 Accelerators

Attach GPUs or TPUs for self hosted models.

05 Networking

Private networking, service mesh and ingress control.

06 ADK deploy

ADK agents deploy to Cloud Run or GKE directly.

HOW IT WORKS · CLOUD RUN & GKE

From request to result

1

Containerise

Package the agent or MCP server.

2

Choose

Cloud Run for simplicity, GKE for control.

3

Deploy

Roll out with revisions or Kubernetes manifests.

4

Scale

Autoscale on load; monitor cost.

WHERE IT SHINES

- Hosting a custom MCP server on Cloud Run.
- Serving an open model with GPUs on GKE.
- Agents needing custom system libraries.

Using it well in production

PRACTITIONER TIPS

1. Default to Agent Runtime or Cloud Run; choose GKE when needed.
2. Use minimum instances for latency critical agents.
3. Isolate untrusted code with sandboxing.

⚠ PITFALLS TO AVOID

1. Running Kubernetes for one small agent.
2. Scale to zero on latency critical paths.
3. Public services without authentication.

CHECK YOURSELF

When would you pick GKE over Agent Runtime?

Answer: When you need custom runtimes, special networking, accelerators or tight control over scheduling.

PRODUCTION & DEPLOYMENT · SERVICE 7

Cloud TPUs & GPUs

Accelerators for tuning, training and serving models.

WHAT IT IS

Google's custom Tensor Processing Units, including the eighth generation announced at Next 2026, plus NVIDIA GPUs, available for training, fine tuning and serving models at many scales.

WHY IT MATTERS

Accelerator choice drives cost and speed of tuning and self hosting. Google offers both custom silicon and GPUs.

LAYER 6 · PRODUCTION



THINK OF IT AS

The engines in the factory, purpose built for AI work.

CAPABILITIES · CLOUD TPUS & GPUS

DEEP DIVE

Six things TPUs and GPUs gives you

01 TPUs

Google designed chips optimised for large model training and inference.

02 GPUs

NVIDIA GPUs for broad framework compatibility.

03 Pods

Scale to thousands of chips for training.

04 Inference serving

High throughput serving for open models.

05 Flexible consumption

On demand, reserved and spot style options.

06 Framework support

JAX, PyTorch and popular serving stacks.

HOW IT WORKS · CLOUD TPUS & GPUS

From request to result

1

Size

Estimate model size, throughput and budget.

2

Select

Pick TPU or GPU type and quantity.

3

Run

Train, tune or serve on the chosen hardware.

4

Optimise

Profile and adjust batch sizes and precision.

WHERE IT SHINES

- Fine tuning Gemma on TPUs.
- Serving Llama on GPUs in GKE.
- Large scale embedding generation.

Using it well in production

PRACTITIONER TIPS

1. Use managed tuning before raw hardware.
2. Profile utilisation; idle chips burn money.
3. Check regional capacity early.

⚠ PITFALLS TO AVOID

1. Over provisioning for peak forever.
2. Ignoring quantisation for inference.
3. Choosing hardware before measuring workload.

CHECK YOURSELF

What should you try before renting raw accelerators for tuning?

Answer: Managed tuning on Agent Platform, which handles hardware and often costs less effort overall.

PRODUCTION & DEPLOYMENT · SERVICE 8

Google Distributed Cloud

Agent Platform for on premises and edge deployments.

WHAT IT IS

Google Cloud hardware and software delivered to your data centre or edge sites, including air gapped options, so Gemini models and agent services run where data must stay.

WHY IT MATTERS

Governments and regulated industries need sovereignty and disconnected operation. GDC brings AI to those environments.

LAYER 6 · PRODUCTION



THINK OF IT AS

A branch office of the cloud built inside your own walls.

CAPABILITIES · GOOGLE DISTRIBUTED CLOUD

DEEP DIVE

Six things Google Distributed Cloud gives you

01 On premises

Runs in your own facilities.

02 Air gapped

Fully disconnected options for classified use.

03 Edge

Low latency AI at factories, stores and sites.

04 Gemini on GDC

Gemini and agent capabilities where permitted.

05 Consistent APIs

Same patterns as public cloud services.

06 Sovereignty

Data and operations remain under local control.

HOW IT WORKS · GOOGLE DISTRIBUTED CLOUD

From request to result

1

Assess

Identify sovereignty and latency needs.

2

Install

Deploy GDC hardware and software.

3

Deploy

Run models and agents locally.

4

Operate

Manage with local or connected operations.

WHERE IT SHINES

A defence agency using Gemini in an air gapped site.

A factory running vision agents at the edge.

A bank meeting strict data residency.

Using it well in production

PRACTITIONER TIPS

1. Confirm which models and features GDC supports.
2. Plan model update processes for disconnected sites.
3. Align with local compliance teams early.

⚠ PITFALLS TO AVOID

1. Assuming every cloud feature exists on GDC.
2. No process for updating air gapped models.
3. Underestimating hardware lead times.

CHECK YOURSELF

Why would an organisation choose Google Distributed Cloud over public cloud regions?

Answer: Sovereignty, air gapped security or edge latency needs that require compute inside their own facilities.

PRODUCTION & DEPLOYMENT · SERVICE 9

Agents CLI

> *Lets coding agents scaffold, evaluate and deploy ADK agents.*

WHAT IT IS

A command line tool for the full agent lifecycle. Developers and coding agents use it to scaffold ADK projects, run evaluations and deploy to Agent Runtime or other targets from scripts and CI.

WHY IT MATTERS

Command line workflows automate easily. The CLI makes agent creation repeatable and lets coding agents build agents too.

LAYER 6 · PRODUCTION



THINK OF IT AS

A command line foreman that assembles agents to specification.

CAPABILITIES · AGENTS CLI

DEEP DIVE

Six things the Agents CLI gives you

01 Scaffold

Generate a structured ADK project from templates.

02 Run locally

Start agents and dev UIs for testing.

03 Evaluate

Execute eval sets from the terminal or CI.

04 Deploy

Push to Agent Runtime, Cloud Run or GKE.

05 Agent friendly

Coding agents can drive it end to end.

06 Scriptable

Fits pipelines and automation.

HOW IT WORKS · AGENTS CLI

From request to result

1

Create

Scaffold a project with chosen template.

2

Develop

Edit code and test locally.

3

Evaluate

Run evals and fix failures.

4

Ship

Deploy from CI with the CLI.

WHERE IT SHINES

- A coding agent scaffolding ten similar agents for regions.
- CI pipelines evaluating every pull request.
- Standardised project layouts across teams.

Using it well in production

PRACTITIONER TIPS

1. Pin CLI versions in CI.
2. Use templates that include evals and observability.
3. Keep deployment config in the repository.

⚠ PITFALLS TO AVOID

1. Deploying from laptops instead of pipelines.
2. Skipping eval steps in scripts.
3. Unreviewed agent generated projects.

CHECK YOURSELF

Why is a CLI valuable when coding agents build agents?

Answer: Coding agents work well with terminal commands, so the CLI lets them scaffold, test and deploy reliably.

PRODUCTION & DEPLOYMENT · SERVICE 10

Cloud Build & Cloud Deploy

> *Build, test, evaluate, security check, deploy, monitor.*

WHAT IT IS

Google Cloud's CI/CD services. Cloud Build runs build, test, evaluation and security steps on every change; Cloud Deploy promotes releases through environments with approvals and rollbacks.

WHY IT MATTERS

Agents change often. A disciplined pipeline makes every change tested, evaluated, secured and reversible.

LAYER 6 · PRODUCTION



THINK OF IT AS

An assembly line with quality checks at every station.

DEEP DIVE

Six things Cloud Build and Deploy gives you

01 Build

Containerise agents and tools reproducibly.

02 Test

Unit and integration tests for tools and logic.

03 Evaluate

Run agent evals as a release gate.

04 Security check

Scan images, dependencies and policies.

05 Progressive deploy

Promote dev to staging to production with approvals.

06 Rollback

Return to the last good release fast.

HOW IT WORKS · CLOUD BUILD & CLOUD DEPLOY

From request to result

1

Commit

A change triggers Cloud Build.

2

Verify

Tests, evals and scans must pass.

3

Promote

Cloud Deploy releases to staging, then production.

4

Monitor

Watch metrics; roll back on regression.

WHERE IT SHINES

- Blocking a prompt change that drops eval scores.
- Canary releases of a new agent version.
- Audit trail of who approved each release.

Using it well in production

PRACTITIONER TIPS

1. Treat prompts and agent configs as code.
2. Make eval thresholds explicit gates.
3. Automate rollback on SLO breach.

⚠ PITFALLS TO AVOID

1. Editing prompts directly in production.
2. No eval stage in the pipeline.
3. Manual deployments without records.

CHECK YOURSELF

Why should a prompt change go through the CI/CD pipeline?

Answer: Prompts change behaviour like code; the pipeline evaluates, secures and records every change before release.

LAYER 6 RECAP

Three things to remember

1. Observe, evaluate, simulate and optimise form a continuous quality loop, not a one time launch gate.
2. Choose compute by need: Agent Runtime for managed agents, Cloud Run or GKE for control, GDC for sovereignty.
3. Ship agents like software: build, test, evaluate, security check, deploy, monitor, every time.

LAYER 6 QUIZ

Test yourself on Production & Deployment

QUESTION 1

Which service stress tests agents with synthetic users before launch?

QUESTION 2

Where would an air gapped government deploy agents?

QUESTION 3

What does Agent Optimizer propose?

QUESTION 4

Name the six pipeline stages on the infographic.

ANSWER KEY, FLIP TO READ

1. Agent Simulation.
2. Google Distributed Cloud.
3. Better instructions from clustered failures.
4. Build, test, evaluate, security check, deploy, monitor.

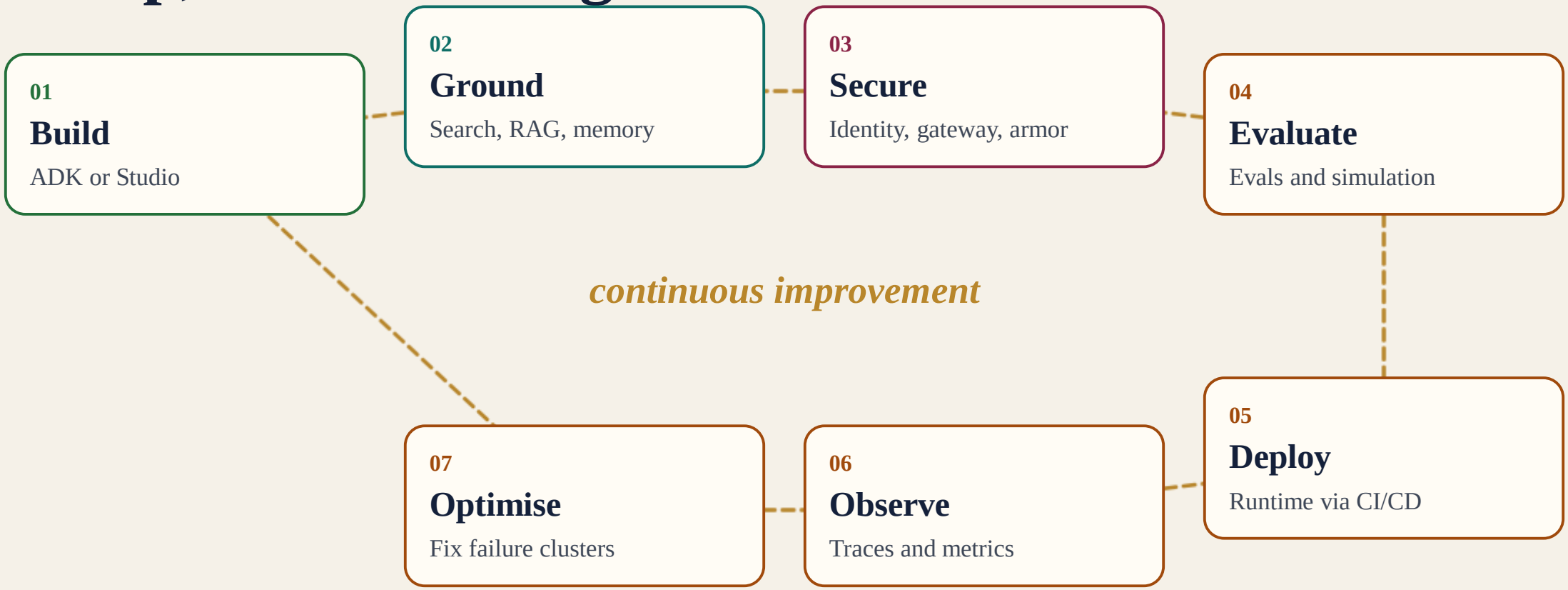
PART SEVEN · PRACTICE

Putting it all together

The lifecycle, three reference architectures, decision guides, a four part hands on lab, checklists, a final quiz and sources.

THE AGENT LIFECYCLE

A loop, never a straight line



REFERENCE ARCHITECTURE

Customer service agent

A retailer answers order, return and refund questions across web chat and voice, escalating to humans when needed.

Enterprise Work & Gemini	Gemini Enterprise app · Inbox approvals	*Staff supervise refunds*
Agent Platform & Tools	ADK · Agent Runtime · Sessions · A2A	*Coordinator plus specialists*
Core Models	Gemini 3.5 Flash · Gemini Live	*Fast text and voice*
Data, Knowledge & Memory	Agent Search · BigQuery · Memory Bank	*Policies, orders, preferences*
Identity, Security & Governance	Agent Identity · Agent Gateway · Model Armor	*Every call governed*
Production & Deployment	Simulation · Evaluation · Observability	*Tested and watched*

REFERENCE ARCHITECTURE

Internal knowledge assistant

A global firm lets employees ask questions across SharePoint, Drive, wikis and tickets, with cited answers and safe actions.

Enterprise Work & Gemini	Gemini Enterprise app · Connectors · Agent Gallery	*One front door*
Data, Knowledge & Memory	Workspace Intelligence · Agent Search · Knowledge Catalog	*Permission aware context*
Core Models	Gemini 3.1 Pro · Grounding	*Deep, cited answers*
Agent Platform & Tools	Agent Designer · MCP	*Team built agents*
Identity, Security & Governance	IAM · VPC Service Controls · Model Armor	*Data stays inside*
Production & Deployment	Monitoring · Agent Evaluation	*Quality over time*

REFERENCE ARCHITECTURE

Data analytics agent

Analysts ask business questions in plain language and get charts, SQL and explanations from governed data.

Data, Knowledge & Memory	BigQuery · Knowledge Catalog · Data Agent Kit	*Certified data and meaning*
Agent Platform & Tools	ADK · MCP · Code Execution	*SQL plus Python sandbox*
Core Models	Gemini 3.1 Pro · Flash Lite	*Plan with Pro, tag with Lite*
Enterprise Work & Gemini	Deep Research · Projects	*Reports to share*
Identity, Security & Governance	Agent Identity · IAM Conditions	*Read only, least privilege*
Production & Deployment	Cloud Build · Cloud Deploy	*Versioned releases*

DECISION GUIDE

Which builder should you use?

Builder	Who	Code needed	Best for
Workspace Studio	Any employee	None	Automating Gmail, Drive, Sheets and Chat tasks
Agent Designer	Business experts	None	Company agents in the Gemini Enterprise app
Agent Studio	Analysts and developers	Low	Visual design, then export to ADK
ADK	Engineers	Full	Complex, multi agent, production systems
Managed Agents API	Product teams	Config	Autonomous agents via REST in a sandbox
Antigravity	Developers	Directing agents	Building software and agents with coding agents

Start as simple as the task allows; graduate when you need control.

DECISION GUIDE

Which model for which job?

Job	Recommended	Why
Hard planning and reasoning	Gemini 3.1 Pro	Deepest reasoning and long context
Most agent steps and coding	Gemini 3.5 Flash	Near Pro agentic quality, lower cost
Routing, tagging, extraction	Gemini 3.1 Flash Lite	Fastest and cheapest tier
Voice and video conversations	Gemini Live API	Real time streaming with barge in
Video, images, music, speech	Veo, Nano Banana, Lyria, TTS	Purpose built media models
Custom style or sovereign serving	Tuned Gemma or open model	Weights you control
Second opinion or preference	Claude via Model Garden	Partner model, same governance

Validate every choice with Agent Platform Evals on your own data.

DECISION GUIDE

Which retrieval or memory service?

Service	Control	Use when
Agent Search	Turnkey	You want Google quality search and grounding fast
RAG Engine	Medium	You need to tune chunking, embeddings and backends
Vector Search	Full	You build custom similarity search at very large scale
BigQuery vector search	Medium	Vectors live next to analytical data you already query
Memory Bank	Managed	You must remember users across sessions
Sessions	Managed	You only need state within one conversation

HANDS ON LAB · PART 1

Prepare the project

1. **Create a project**
   Use a fresh Google Cloud project with billing and the Agent Platform API enabled.

2. **Pick a region**
   Choose one where Runtime, Sessions and Memory Bank are offered, such as London.

3. **Set identities**
   Plan an agent identity with least privilege roles from day one.

PYTHON SETUP SKETCH

```
# environment settings
PROJECT_ID = "your_project"
LOCATION = REGION_ID # e.g. London

# install ADK with pip, then:
import vertexai
vertexai.init(project=PROJECT_ID,
location=LOCATION)
```

HANDS ON LAB · PART 2

Build the agent with ADK

1. **Write a tool**
   A plain Python function with a clear docstring becomes a tool.

2. **Define the agent**
   Give it a model, a short instruction and the tool.

3. **Keep it small**
   One job per agent; add sub agents later.

AGENT.PY

```
from google.adk.agents import Agent

def order_status(order_id: str):
# look up the order in your system
return ORDERS.get(order_id, "not found")

root_agent = Agent(
name="order_helper",
model=MODEL_ID, # a Gemini Flash model
instruction="Answer order questions.",
tools=[order_status],
)
```

HANDS ON LAB · PART 3

Test and evaluate locally

1. **Chat in the dev UI**
   Run the ADK web UI and inspect every event and tool call.

2. **Write an eval set**
   Ten real questions with expected tool calls and answers.

3. **Gate on scores**
   Fail the build if tool accuracy or answer quality drops.

EVALUATION SKETCH

```
# eval case, written as JSON
{
"query": "Where is order 1042?",
"expected_tool": "order_status",
"reference": "Order 1042 has shipped."
}

# run the ADK eval command in CI
```

HANDS ON LAB · PART 4

Deploy, govern and publish

1. **Deploy to Runtime**
   Ship through Cloud Build to Agent Runtime in your region.

2. **Register and protect**
   Add it to Agent Registry, set Gateway policy and Model Armor.

3. **Publish to people**
   Surface it in the Gemini Enterprise app and watch traces.

DEPLOYMENT SKETCH

```
# deploy sketch
from vertexai import agent_engines

remote = agent_engines.create(
agent_engine=root_agent,
display_name="order helper",
)
print(remote.resource_name)
```

CHECKLIST

Security before go live

- [ ] **1.** Every agent has its own identity and owner.
- [ ] **2.** Roles follow least privilege and use groups.
- [ ] **3.** All tool and agent traffic routes through Agent Gateway.
- [ ] **4.** Model Armor screens prompts, tool results and outputs.
- [ ] **5.** Agents and MCP servers are in Agent Registry.
- [ ] **6.** VPC Service Controls perimeter covers data services.
- [ ] **7.** CMEK is enabled where policy requires it.
- [ ] **8.** Write actions need human approval in Inbox.
- [ ] **9.** Agent logs feed Security Command Center and SecOps.
- [ ] **10.** Red team simulation passed, including injection.

CHECKLIST

Production readiness

- [ ] **1.** Golden eval set exists and gates every release.
- [ ] **2.** Simulation covered angry users and tool failures.
- [ ] **3.** Tracing is on with sensitive data redacted.
- [ ] **4.** SLOs and alerts are defined with owners.
- [ ] **5.** Cost alerts and step limits are set.
- [ ] **6.** Provisioned or batch consumption is right sized.
- [ ] **7.** Rollback is tested in Cloud Deploy.
- [ ] **8.** Region supports every service used.
- [ ] **9.** Users can view and delete their memories.
- [ ] **10.** A named owner reviews quality monthly.

ECONOMICS

Six levers that control agent cost

Tier routing

Send simple steps to Flash Lite and reserve Pro for hard reasoning.

Context caching

Cache long stable prefixes such as manuals and policies.

Batch

Move offline jobs to discounted batch inference.

Provisioned baseline

Reserve steady traffic; let spikes overflow.

Retrieve, do not stuff

Fetch the right pages instead of filling the context window.

Watch the loops

Cap agent steps and alert on token spikes per session.

Example list prices at time of writing: 3.1 Flash Lite about $0.25 input and $1.50 output per million tokens; 3.1 Pro from about $2 and $12. Always check current pricing.

FINAL QUIZ, PART ONE

The whole stack, round one

QUESTION 1

Which product did Vertex AI evolve into?

QUESTION 2

What was Agent Engine renamed to?

QUESTION 3

Which layer holds Memory Bank?

QUESTION 4

Which open protocol links agents to tools?

ANSWER KEY, FLIP TO READ

1. Gemini Enterprise Agent Platform.
2. Agent Runtime.
3. Layer 4, Data, Knowledge and Memory.
4. MCP.

FINAL QUIZ, PART TWO

The whole stack, round two

QUESTION 1

Name the four platform pillars.

QUESTION 2

Which service clusters failures and suggests fixes?

QUESTION 3

Where do employees run partner agents?

QUESTION 4

What screens tool results for hidden instructions?

ANSWER KEY, FLIP TO READ

1. Build, scale, govern, optimize.
2. Agent Optimizer.
3. Agent Gallery in the Gemini Enterprise app.
4. Model Armor.

THE WHOLE STACK

Fifty two services on one slide

Production & Deployment	Agent Observability, Agent Evaluation, Agent Simulation, Agent Optimizer, Cloud Monitoring & Logging, Cloud Run & GKE, Cloud TPUs & GPUs, Google Distributed Cloud, Agents CLI, Cloud Build & Cloud Deploy
Identity, Security & Governance	Agent Identity, Agent Registry, Agent Gateway, Model Armor, Security Command Center, Wiz, Google Security Operations, IAM & VPC Service Controls
Data, Knowledge & Memory	BigQuery, Knowledge Catalog, Workspace Intelligence, Data Agent Kit, Agent Search, RAG Engine, Vector Search, Memory Bank
Enterprise Work & Gemini	Gemini Enterprise App, Agent Designer & Skills, Agent Gallery, Deep Research & Data Insights, Workspace Studio, Projects, Canvas & Inbox, Connectors & BYO MCP, Antigravity
Agent Platform & Tools	Agent Development Kit, Agent Studio, Agent Garden, Managed Agents API, Agent Runtime, Sessions & Code Execution, A2A Protocol, MCP
Core Models	Gemini Models, Veo, Nano Banana & Lyria, Anthropic Claude, Gemma, Llama & Mistral, Gemini Embedding & Live, Model Garden, Agent Platform Evals, Model Registry, Grounding with Google Search, Provisioned Throughput

SOURCES AND FURTHER READING

Where to verify and go deeper

01 Introducing Gemini Enterprise Agent Platform, Google Cloud Blog

02 Agent Platform release notes and naming changes, Google Cloud Docs

03 Scale your agents: Runtime, Sessions, Memory Bank, Code Execution

04 Developer guide to Gemini 3.5 Flash, Google Cloud Docs

05 Transition from Dataplex Universal Catalog to Knowledge Catalog

06 Workspace Studio launch, Google Workspace Updates

07 Google Cloud Next 2026 analysis, Bain and Company

08 Google claims answers for agent sprawl, The Register

09 Google Cloud launches Agentic Data Cloud, ChannelLife

Structure based on the infographic by Prashant Rathi. Each title links to its source.

END OF TUTORIAL

Now build one agent, end to end.

Pick one real task, one Gemini tier, one data source and one evaluation set. Ship it through the full stack, then come back to this deck layer by layer.

build · scale